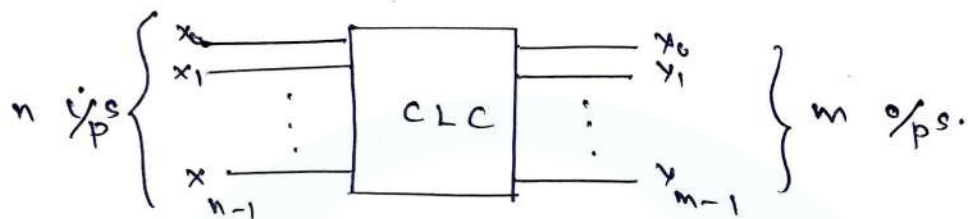


Module 4

COMBINATIONAL LOGIC CIRCUIT (CLC).

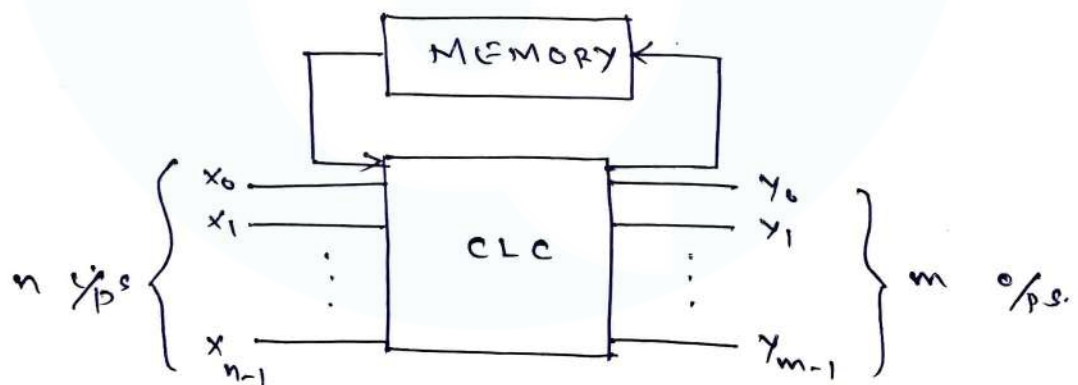
The present $\%_p$ depends only on present $\%_p$.

eg:- Adder, Subtractor, ENL, DEC, Mux, DEMux.



SEQUENTIAL LOGIC CKT (SLC).

The present $\%_p$ depends on present $\%_p$ and past $\%_p$. In other words one present $\%$ depends on present $\%$ and past $\%$.



Features of SLC.

- (1) The network has a memory to store the past $\%_p$.
- (2) There are one or more feed back paths from $\%_p$ to $\%_p$.

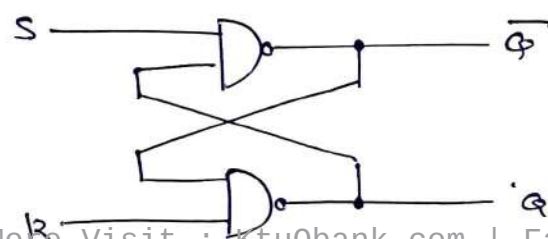
- (3) The operation of SLC may or may not depends on clock.
- (4) SLC remain in a given logic state even after the c/p's have been removed.
- (5) SLC's are more complex than CLC's.
- (6) SLC's are constructed using CLC's.

FLIP - FLOP.

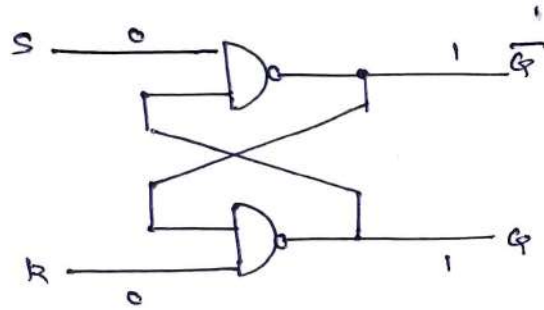
Basic memory unit to store 1 bit (single bit)
 The most fundamental circuit in the group of flip-flop is SR-latch.

SR LATCH.

The basic SR latch has two NAND gates connected back-to-back. The SR latch has two external c/p's S and R, and two feed-back c/p's Q and $\bar{Q}$. The present state of the SR latch is Q_0 . After one clock pulse, one o/p will change to the next state Q_1 .

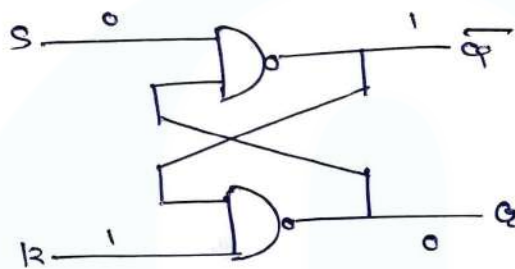


Case 1: $S=0$, $R=0$



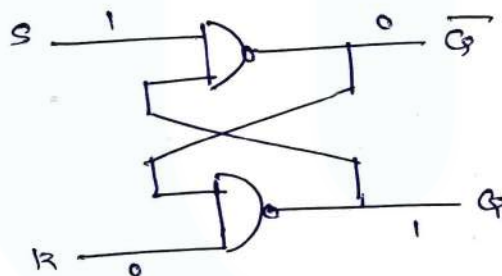
$Q=1$; $\bar{Q}=1$
(Not permitted)

Case 2: $S=0$, $R=1$



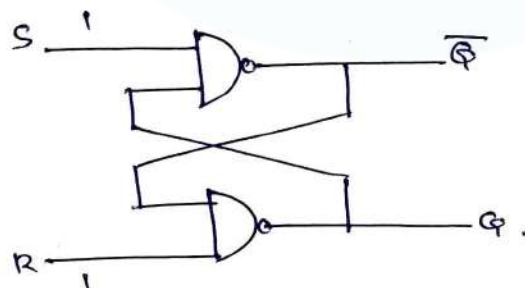
$Q=0$; $\bar{Q}=1$
(Reset).

Case 3: $S=1$, $R=0$



$Q=1$; $\bar{Q}=0$
(Set).

Case 4: $S=1$, $R=1$

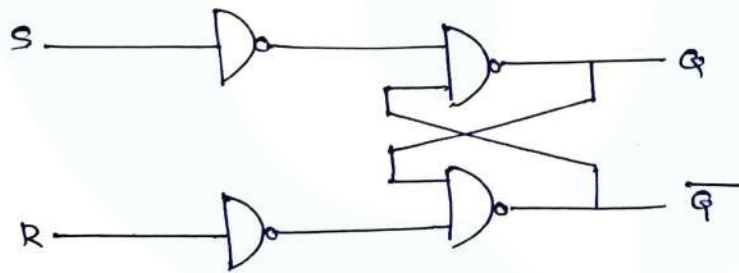


If present state $Q_0=0$, $\bar{Q}=1$ and $Q=0$.

If present state $Q_0=1$; $\bar{Q}=0$ and $Q=1$. This means the next state is same as the present state.

Truth Table.

S	R	Q_0	Q_1
0	0	0	Not permitted.
0	0	1	
0	1	0	0 (Reset)
0	1	1	
1	0	0	1 (Set)
1	0	1	
1	1	0	0 } Q_0 .
1	1	1	

S R Flip-Flop.

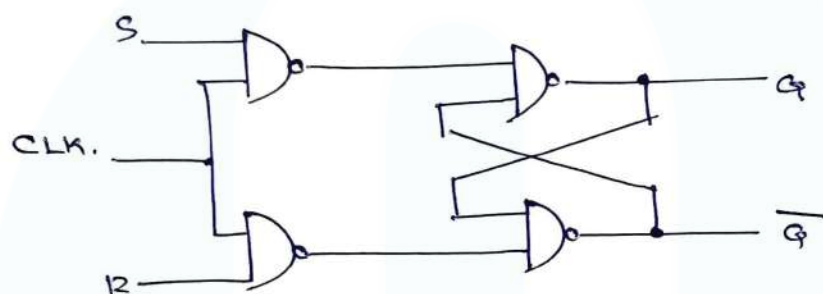
S	R	Q_0	Q_1
0	0	0	0 } Q_0
0	0	1	
0	1	0	0 } Reset
0	1	1	
1	0	0	1 } Set
1	0	1	
1	1	0	Not permitted
1	1	1	

Here, one state 1-1 is forbidden. This is an asynchronous sequential logic circuit, as it does not depend on a clock for its working.

An SR flip-flop can be converted into a synchronous sequential logic ckt. The ckt will be enabled, if and only if clock $\bar{c}/p$ $clk=1$.

If $clk=0$, both $\bar{c}/p$ NAND gates are disabled and the ckt will not function at all.

When $clk=1$, both the $\bar{c}/p$ NAND are enabled and the ckt will function as a normal SR FF.



CLK	S	R	Q_0	Q_1
0	← circuit Disabled →			
1	0	0	0/1	0/1
1	0	1	0/1	0
1	1	0	0/1	1
1	1	1	0/1	Not permitted.

JK Flip-Flop.

Desired truth table of JK FF.

J	K	Q_0	Q_1
0	0	Q_0	Q_0
0	1	0/1	0
1	0	0/1	1
1	1	$\bar{Q}_0$	$\bar{Q}_0$

Conversion of SR FF into JK FF.

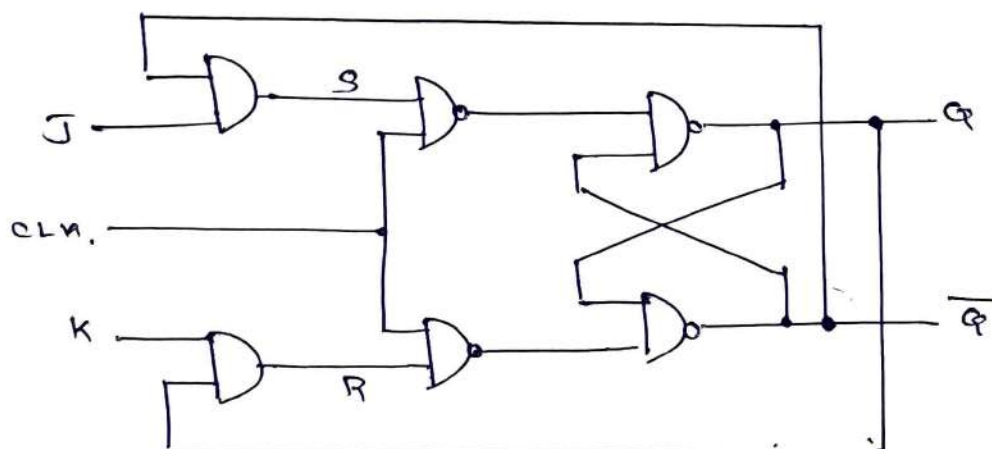
J	K	Q_0	Q_1	S	R
0	0	0	→ 0	0	0
0	0	1	→ 1	0	0
0	1	0	→ 0	0	0
0	1	1	→ 0	0	1
1	0	0	→ 1	1	0
1	0	1	→ 1	0	0
1	1	0	→ 1	1	0
1	1	1	→ 0	0	1

JK	Q_0	$\overline{Q_0}$
$\overline{J}\overline{K}$		
$\overline{J}K$		
$J\overline{K}$	1	
JK	1	

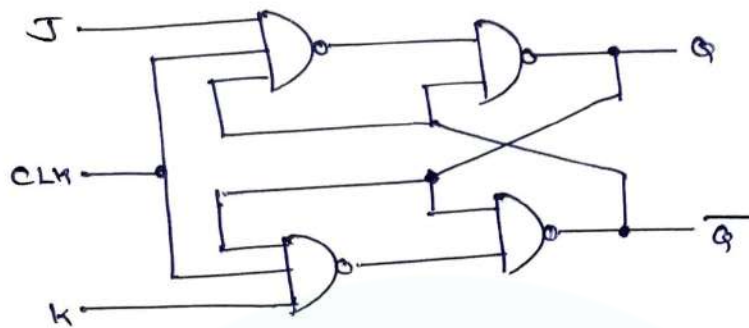
$$S = J \overline{Q_0}$$

JK	Q_0	$\overline{Q_0}$
$\overline{J}\overline{K}$		
$\overline{J}K$		
$J\overline{K}$	1	
JK	1	

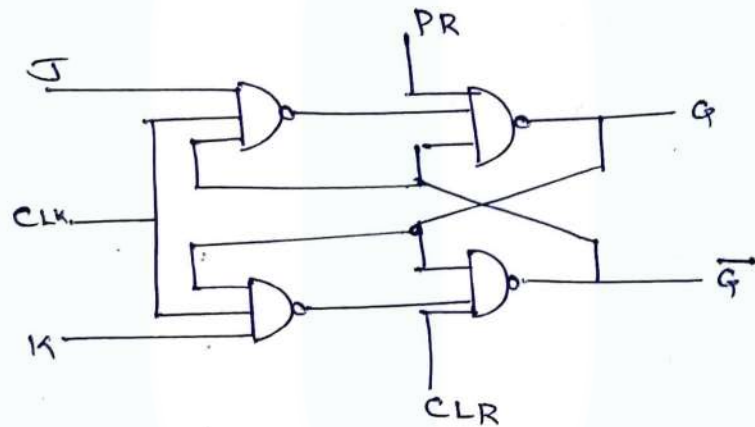
$$R = K Q_0.$$



Simplified Circuit Diagram for JK FF



JK FF with Preset & Clear i/p



The preset and clear inputs do not obey the clock pulse, and override them in action. Hence they are called the asynchronous i/p's.

Let $PR=0$ and $CLR=1$. Then, as and when PR is made 0, Q will jump to 1 from whatever state it is in.

When $PR=1$ and $CLR=0$, the lower NAND produces 1 at its o/p $\bar{Q}$; $\bar{Q}=1$. This makes $Q=0$. Thus whatever be the state, Q will '0'.

When both $PR = CLR = 1$, the NANDs are enabled and the JK FF functions in the normal way.

The state $PR = CLR = 0$ is forbidden, since both $Q = \overline{Q}$ will be 1 in this state.

PR	CLK	Q_0	Q_1
0	0	← Forbidden →	
0	1	0/1	1
1	0	0/1	0
1	1	← Normal operation →	

D - FLIP FLOP. / DELAY FF

This FF has only one c/p , called Delay or D c/p . When D is 0, Q will become 0 after the propagation delay t_p ; similarly, if $D=1$, c/p will become 1, irrespective of the previous value of the c/p . The FF will always produce an c/p equal to the c/p , after a time-delay equal to t_p . Since the circuit c/p is produced after a time-delay, this is called the delay FF (D-FF).

D	Q_0	Q_1
0	0	0
0	1	0
1	0	1
1	1	1

T-Flip-Flop (Toggle FF)

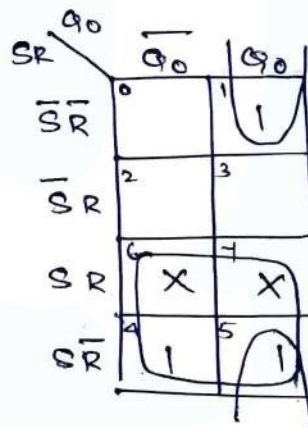
When $T=0$, $Q_1 = Q_0$. However $T=1$ $Q_1 = \overline{Q_0}$

T	Q_0	Q_1
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic Equation of flip-flops.

SR FF.

S	R	Q_0	Q_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X



$$\underline{\underline{Q_1 = S + \overline{R} Q_0}}$$

$$\underline{\underline{Q(t+1) = S(t) + \overline{R(t)} \cdot Q(t)}}$$

J K FF

J	K	Q ₀	Q ₁
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK	Q ₀	Q ₁
$\bar{J}\bar{K}$		1
$\bar{J}K$		
$J\bar{K}$	1	
JK	1	1

$$Q_1 = J\bar{Q}_0 + \bar{K}Q_0$$

D FF

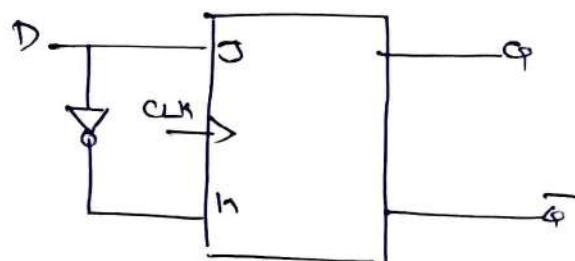
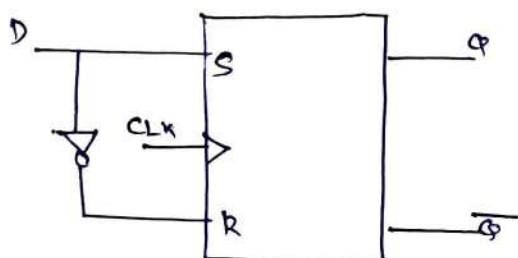
$$Q_1 = D$$

D	Q ₀	Q ₁
0	0	0
0	1	0
1	0	1
1	1	1

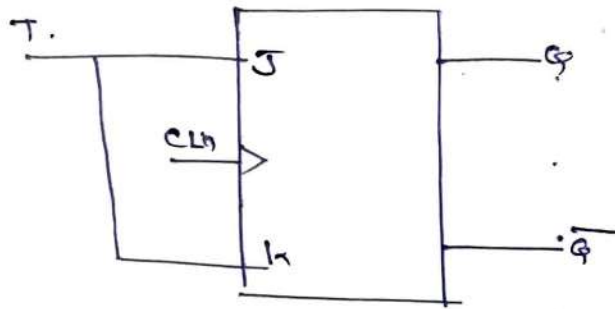
T FF

$$Q_1 = T \oplus Q_0$$

T	Q ₀	Q ₁
0	0	0
0	1	1
1	0	1
1	1	0

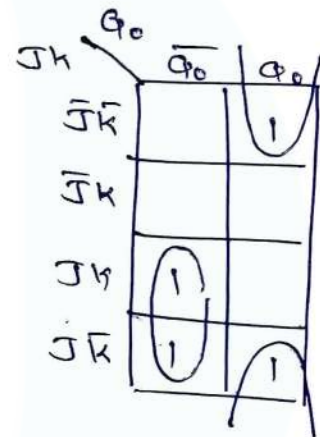
D-FF using SR & JK FF

T FF using JK FF.



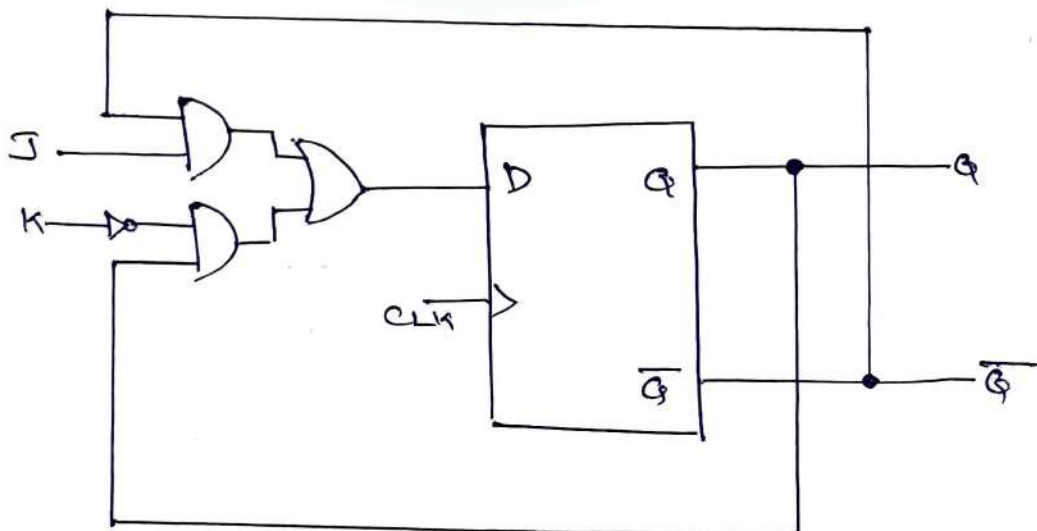
Conversion of D-FF into JK FF.

J	K	Q_0	Q_1	D
0	0	0	→ 0	0
0	0	1	→ 1	1
0	1	0	→ 0	0
0	1	1	→ 0	0
1	0	0	→ 1	1
1	0	1	→ 1	1
1	1	0	→ 1	1
1	1	1	→ 0	0



$$\underline{\underline{D = J \bar{Q}_0 + \bar{K} Q_0}}$$

1

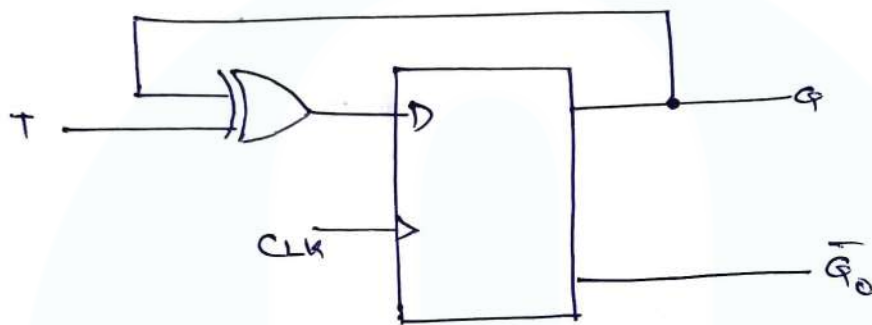


Conversion of D-FF into T-FF

T	Q ₀	Q ₁	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

$$D = \overline{T} Q_0 + T \overline{Q_0}$$

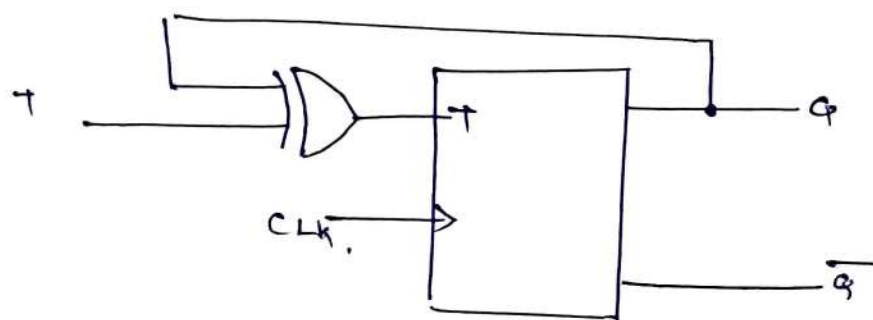
$$\underline{\underline{D = T \oplus Q_0}}$$



Conversion of T-FF to D-FF.

D	Q ₀	Q ₁	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$$\underline{\underline{T = D \oplus Q}}$$

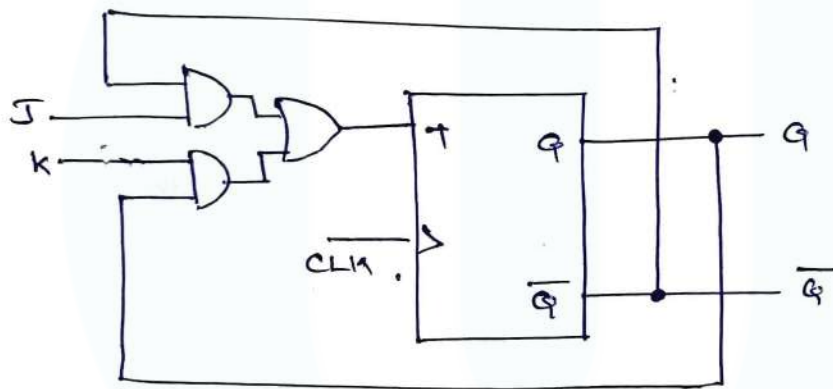


CONVERSION of T-FF to JK FF

J	k	Q ₀	Q ₁	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀
Jk	Q ₀	Q ₀

$$T = J\bar{Q}_0 + KQ_0$$



EXCITATION TABLES.

S	R	Q ₀	Q ₁
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Q ₀	Q ₁	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

JK FF

J	K	Q_0	Q_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Q_0	Q_1	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

D-FF

D	Q_0	Q_1
0	0	0
0	1	0
1	0	1
1	1	1

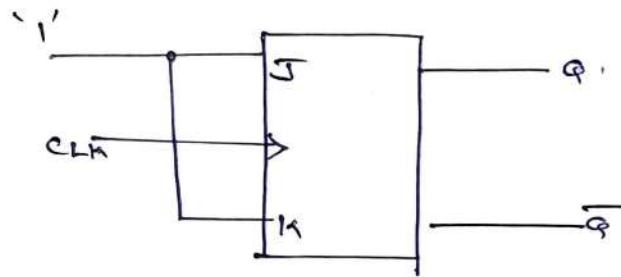
Q_0	Q_1	D
0	0	0
0	1	1
1	0	0
1	1	1

T-FF

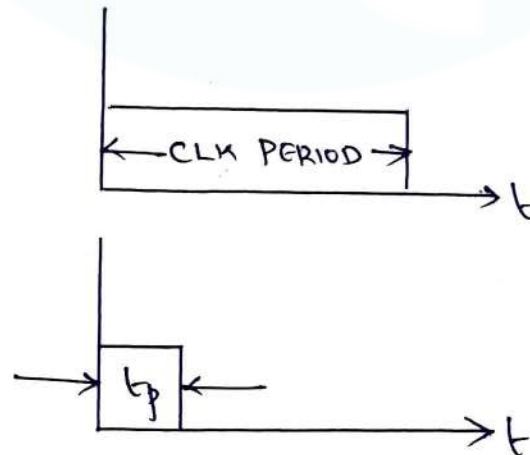
T	Q_0	Q_1
0	0	0
0	1	1
1	0	1
1	1	0

Q_0	Q_1	T
0	0	0
0	1	1
1	0	1
1	1	0

RACE AROUND CONDITION.



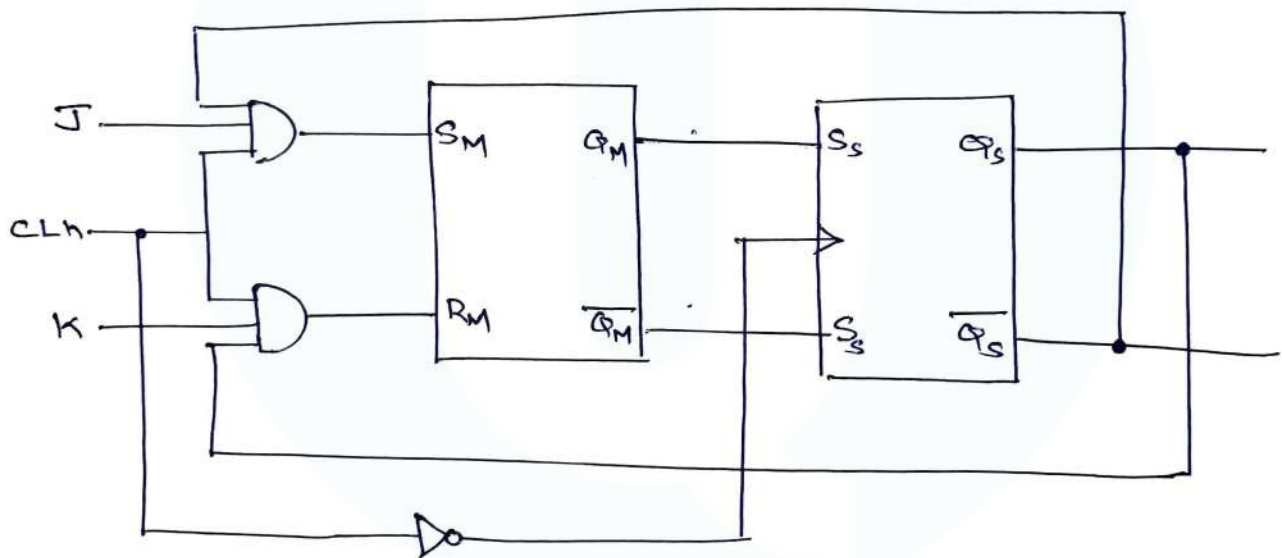
Let us initially assume that $J = K = 1$ and $Q_0 = 0$. When the clk is applied Q_1 becomes 1, the o/p jumps from 0 to 1. This change in o/p from 0 to 1 occurs in a duration of $t = t_p$ (propagation time). Now because of the feed back paths from $\bar{Q}$ to J and Q to K, the situation becomes $J = K = 1$ and $Q_0 = 1$. If clk is still on Q_1 becomes '0'. Usually CLK's have large time duration compared to the propagation delay of the gates.



When $J = K = 1$ and $clk = 1$, the o/p will toggled from 0 to 1 and 1 to 0 continuously until the clk pulses becomes zero. The result of this toggling action is that when the clk becomes zero, we will not know in what stage it would be. This is called race around condition.

Remedy to the Race Around Condition.

Master - Slave JK FF.



Here two SR FF are in JK mode of operation. First FF is the master and 2nd one is the slave. The slave c/p's are connected to the o/p of the master. The master is driven by clock pulse CLK , while the slave is driven by $\overline{CLK}$.

the feed back connection can be seen to be from slave o/p to master i/p's.

When clk is on, the master gets enabled and slave gets disabled so the i/p's at the terminals S_M and R_M of the master will appear at its o/p's Q_M and $\overline{Q_M}$ respectively. i.e., the external i/p's appeared at the i/p's S_S and R_S of the slave.

When $CLK=0$, the master is disabled and slave is enabled (because $\overline{CLK}=1$). Thus the i/p's S_S and R_S will appear at Q_S and $\overline{Q_S}$ respectively. Since master is disabled, these o/p will not appear at S_S and R_S . Hence no feed back problems occurs.

BINARY COUNTERS.

The basic FF used for counter is T-FF.

Triggering.

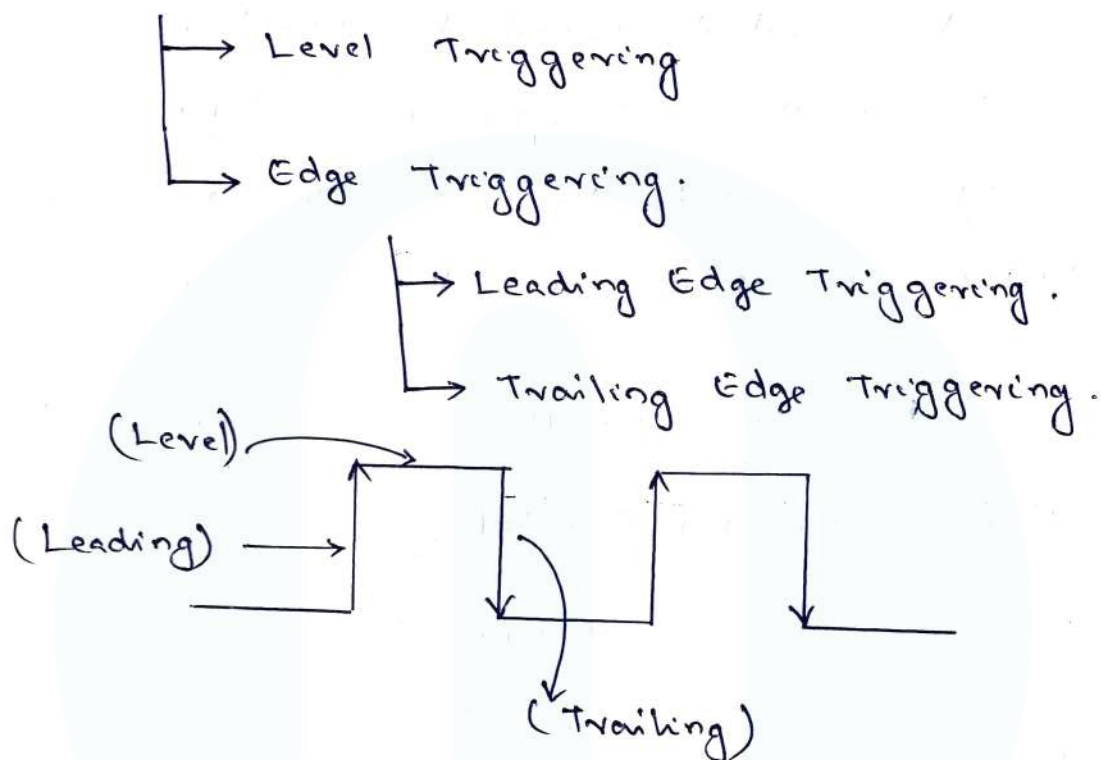
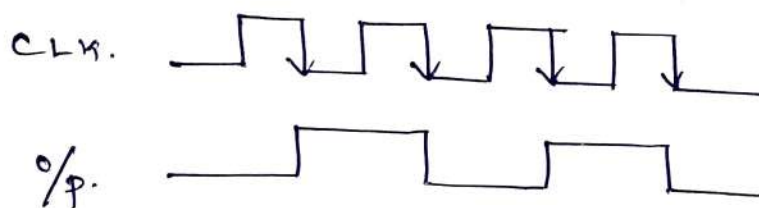
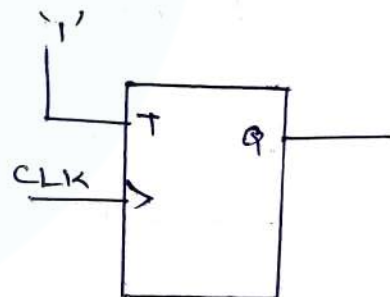


Figure shows, a T-FF with i/p $T=1$ and continuous edge is triggered. $\therefore$ High-to-low and low-to-high trailing edge of the CLK pulses as shown in figure.



From the above fig. it is shown that, the clk divides the i/p pulse freq. by a factor of two.

By cascading the required number of T-FFs we can then divide the i/p freq by any desired factor. This is a method of counting the number of i/p pulses. Such a ckt, which can be used to count the number of i/p pulses, is known as a frequency counter. Since we are using the circuit to count the number of binary pulses, we can also call it a binary counter.

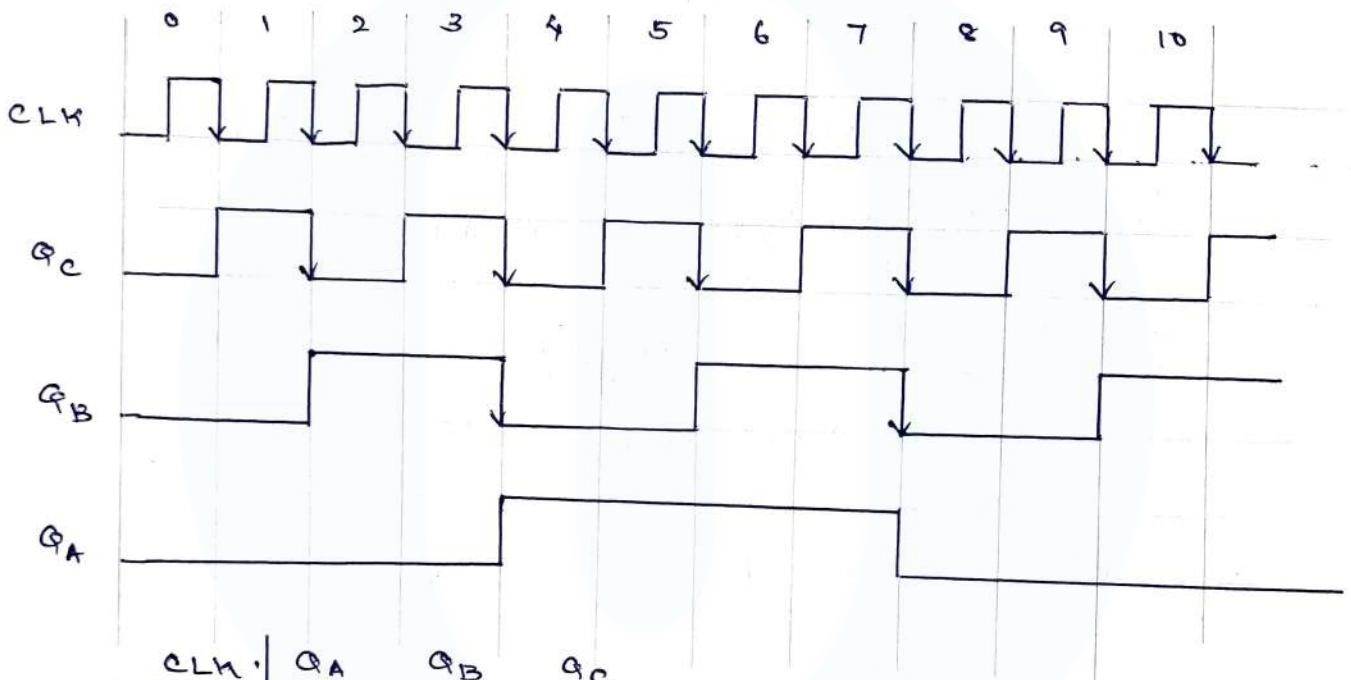
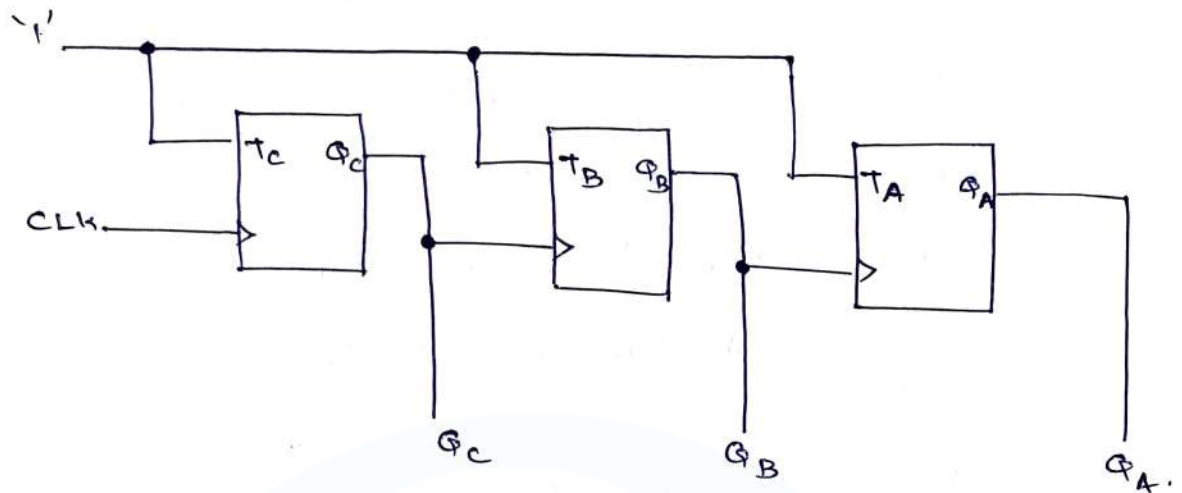
Binary counters are mainly classified into.

- (1) Asynchronous Counter.
- (2) Synchronous Counter.

ASYNCHRONOUS COUNTER.

An asynchronous counter is one whose FFs do not depend on a clock-pulse transition for changing their states. Instead, they depend on the o/p's from their previous state to change their present state.

3-bit asynchronous counter.



CLK	Q_A	Q_B	Q_C
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0
9	0	0	1
10	0	1	0

Using the counter clk the number of incoming pulses can be counted. The major disadvantage of the ripple counter is the large time delay in completing the counting process. The counting will be completed only after all the $\frac{1}{2}$ pulses have rippled through all the cascaded flip-flops.

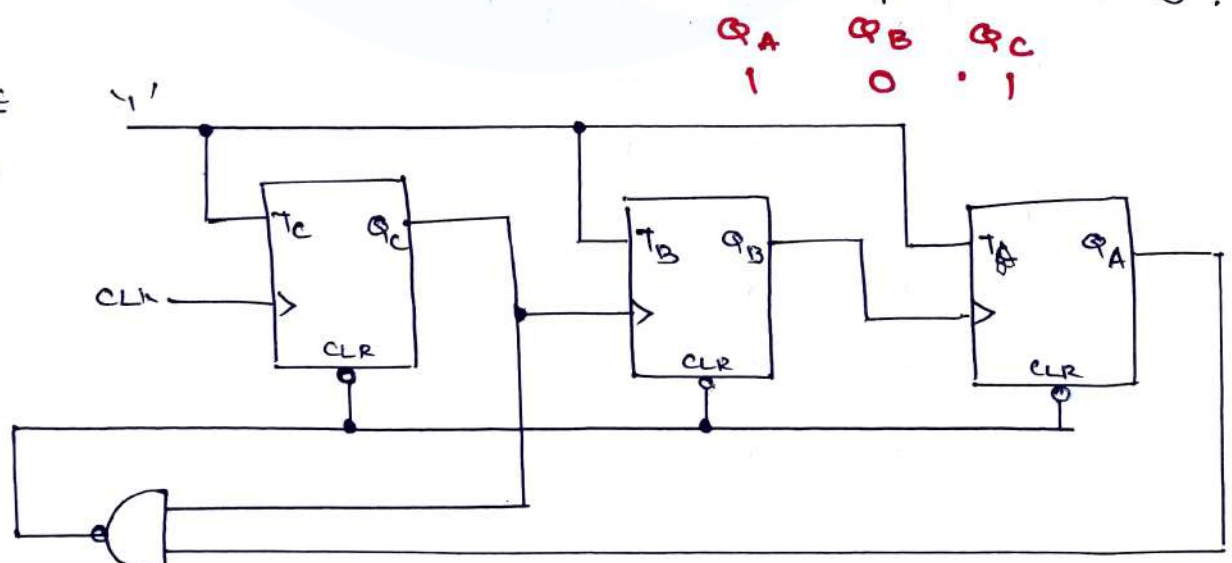
Modulo 2^n counter or Mod 2^n counter.

The counter which can count from 0 to $2^n - 1$. (n - no. of FF used).

eg:- A modulo 8 counter or modulo 2^3 counter can count from 0 to 7.

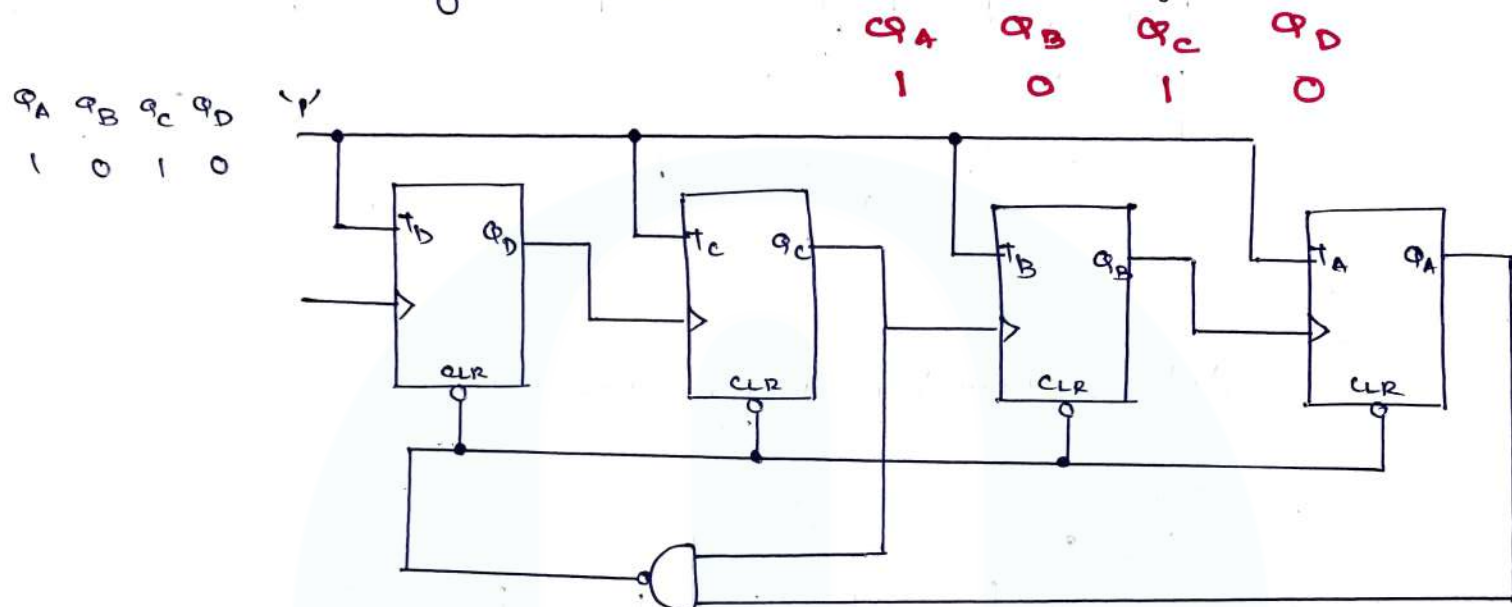
Modulo - 5 counter.

A modulo - 5 counter can count from 0 to 4. The min no. of FF required is 3.



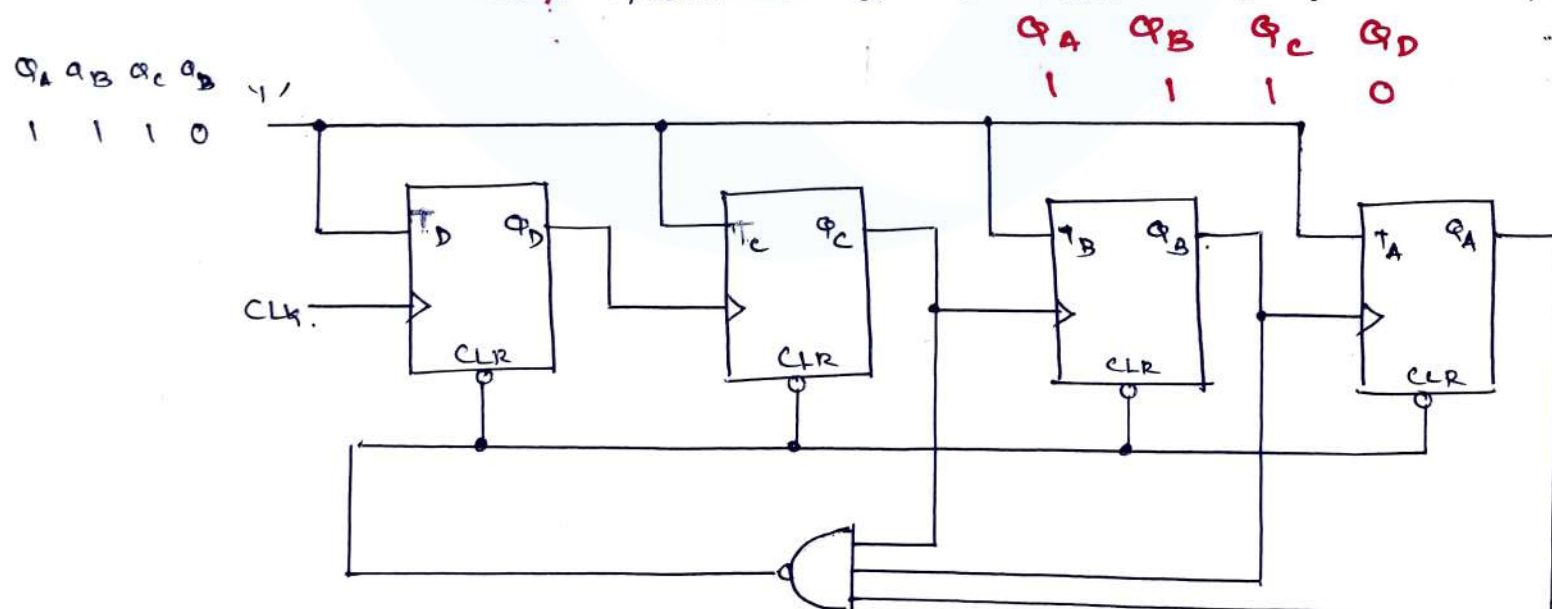
Modulo - 10 or Decade Counter.

Counts from 0 to 9. This can be constructed from a modulo-16 counter by resetting the counter at the tenth pulse.

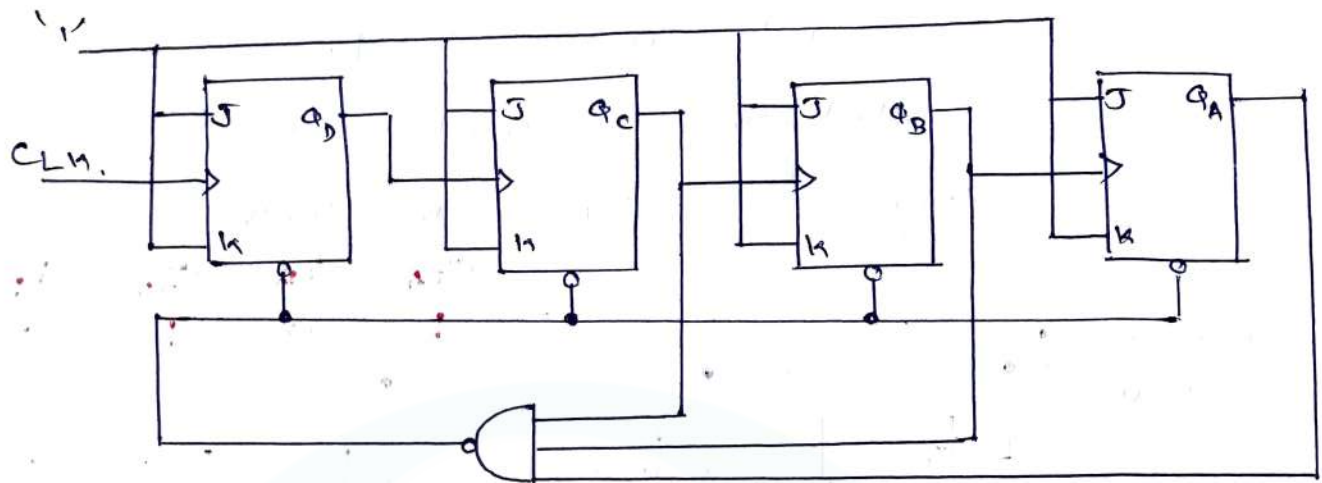


Modulo - 14 Counter.

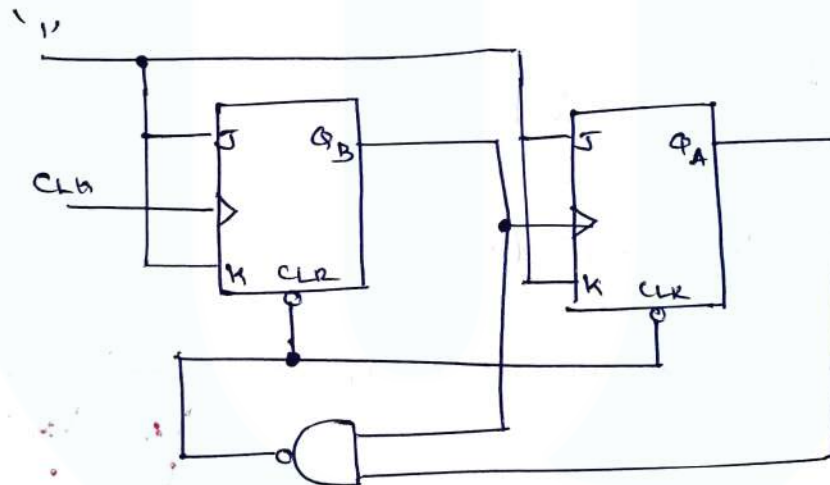
Counts from 0 to 13. Min no. of T-FF = 4.



Using JK-FF



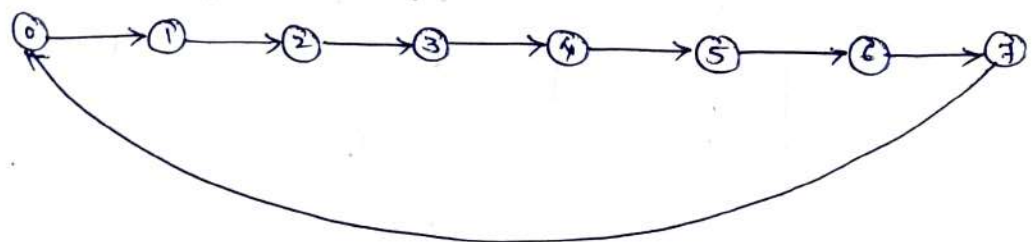
Module-3 counter using JK-FF.



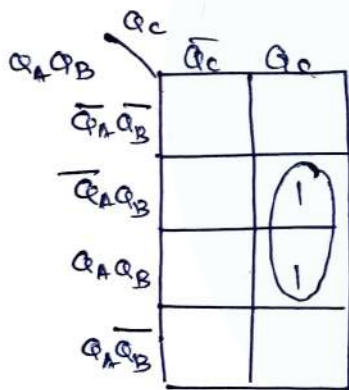
Synchronous Counter	Asynchronous Counter.
(1) Propagation delay time is much less than ripple counter.	Propagation delay time is very large.
(2) Freq. of operation is very high.	Very low.
(3) Extra logic gates required is high.	Minimum extra logic required.
(4) High cost.	Low cost.
(5) Easy to design.	Design based on experience only.
(6) std logic package available.	std logic packages are not available.

Design of Synchronous counters.

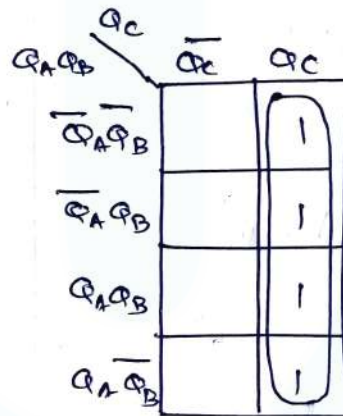
Q:- Design a 3-bit synchronous counter using T-FF.



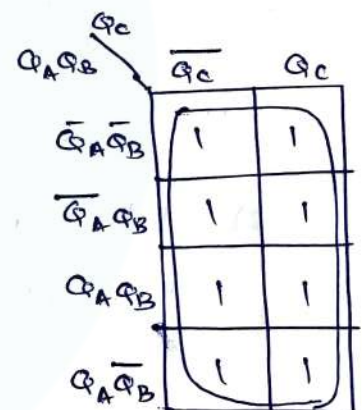
	PS			NS			FF $\div P^S$		
	Q_A	Q_B	Q_C	Q_A	Q_B	Q_C	T_A	T_B	T_C
0	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
2	0	1	0	0	1	1	0	0	1
3	0	1	1	1	0	0	1	1	1
4	1	0	0	1	0	1	0	0	1
5	1	0	1	1	1	0	0	1	1
6	1	1	0	1	1	1	0	0	1
7	1	1	1	0	0	0	1	1	1



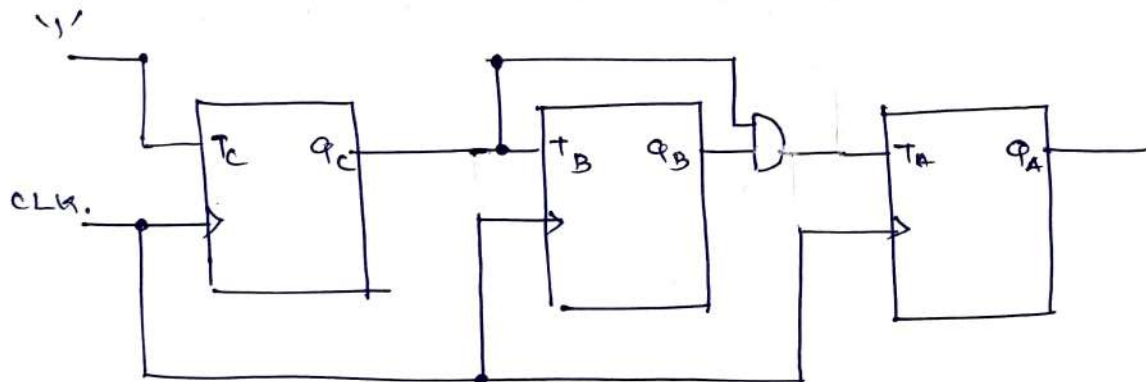
$$T_A = Q_B Q_C$$



$$T_B = Q_C$$



$$T_C = 1$$



Q:- Design a 4-bit Synchronous Counter using T-FF

	PS				NS				T-FF $\frac{Q}{PS}$			
	Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1	1
2	0	0	1	0	0	0	1	1	0	0	0	1
3	0	0	1	1	0	1	0	0	0	1	1	1
4	0	1	0	0	0	1	0	1	0	0	0	1
5	0	1	0	1	0	1	1	0	0	0	1	1
6	0	1	1	0	0	1	1	1	0	0	0	1
7	0	1	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	1	0	0	1	0	0	0	1
9	1	0	0	1	1	0	1	0	0	0	1	1
10	1	0	1	0	1	0	1	1	0	0	0	1
11	1	0	1	1	1	1	0	0	0	1	1	1
12	1	1	0	0	1	1	0	1	0	0	0	1
13	1	1	0	1	1	1	1	0	0	0	1	1
14	1	1	1	0	1	1	1	1	0	0	0	1
15	1	1	1	1	0	0	0	0	1	1	1	1

T_3

Q_3, Q_2	Q_1, Q_0	00	01	11	10
00					
01				1	
11				1	
10					

T_2

Q_3, Q_2	Q_1, Q_0	00	01	11	10
00				1	
01				1	
11				1	
10				1	

T_1

Q_3, Q_2	Q_1, Q_0	00	01	11	10
00		1	1		
01		1	1		
11		1	1		
10		1	1		

T_0

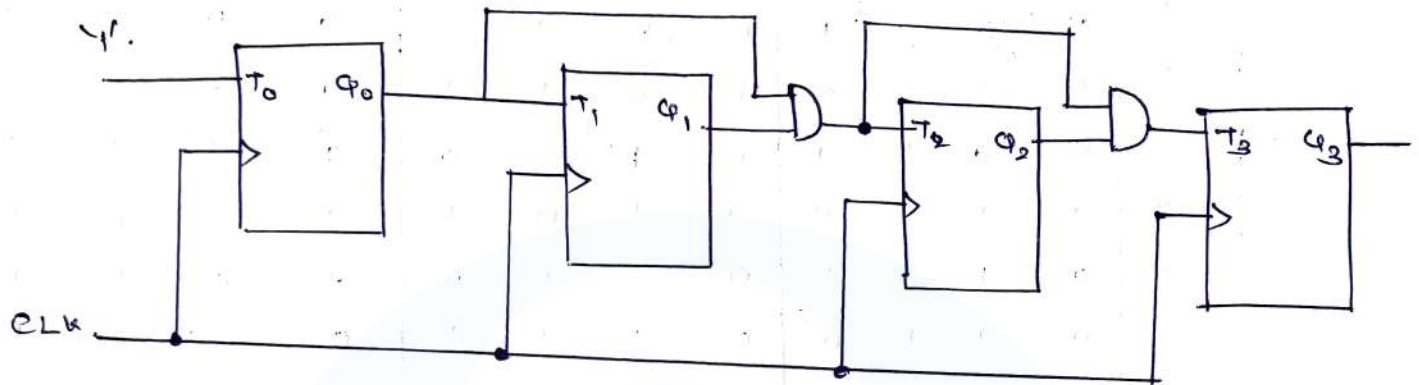
Q_3, Q_2	Q_1, Q_0	00	01	11	10
00	1	1	1	1	
01	1	1	1	1	
11	1	1	1	1	
10	1	1	1	1	

$$T_3 = Q_2 Q_1 Q_0$$

$$T_2 = Q_1 Q_0$$

$$T_1 = Q_0$$

$$T_0 = 1$$



Q:- Design a synchronous counter (decade) using JK FF.

Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

$$Q_3 Q_2 \begin{array}{c} Q_1 Q_0 \\ \hline \bar{Q}_1 \bar{Q}_0 & \bar{Q}_1 Q_0 & Q_1 \bar{Q}_0 & Q_1 Q_0 \end{array}$$

$\bar{Q}_3 \bar{Q}_2$				
$\bar{Q}_3 Q_2$			1	
$Q_3 \bar{Q}_2$	x	x	x	x
$Q_3 Q_2$	x	x	x	x

$$J_3 = Q_2 Q_1 Q_0$$

$$Q_3 Q_2 \begin{array}{c} Q_1 Q_0 \\ \hline \bar{Q}_1 \bar{Q}_0 & \bar{Q}_1 Q_0 & Q_1 \bar{Q}_0 & Q_1 Q_0 \end{array}$$

$\bar{Q}_3 \bar{Q}_2$	x	x	x	x
$\bar{Q}_3 Q_2$	x	x	x	x
$Q_3 \bar{Q}_2$	x	x	x	x
$Q_3 Q_2$		1	x	x

$$K_3 = Q_0$$

		1	
x	x	x	x
x	x	x	x
		x	x

$$J_2 = Q_1 Q_0$$

x	x	x	x
		1	
x	x	x	x
x	x	x	x

$$K_2 = Q_1 Q_0$$

	1	x	x
	1	x	x
x	x	x	x
		x	x

$$J_1 = \bar{Q}_3 Q_2$$

x	x	1	
x	x	1	
x	x	x	x
x	x	x	x

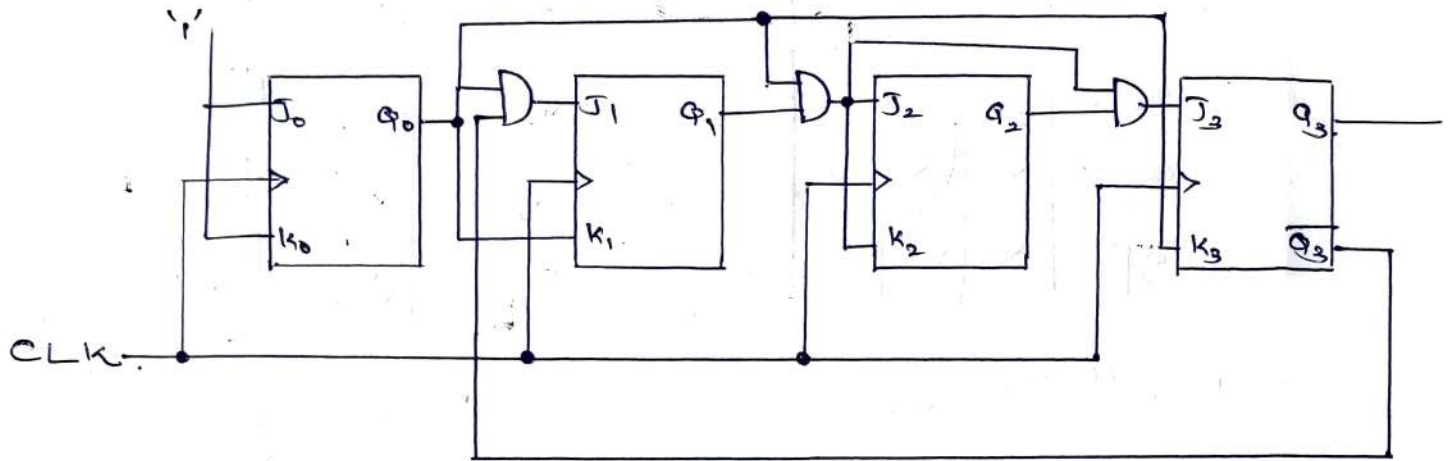
$$K_1 = Q_0$$

1	x	x	1
1	x	x	1
x	x	x	x
1	x	x	x

$$J_0 = 1$$

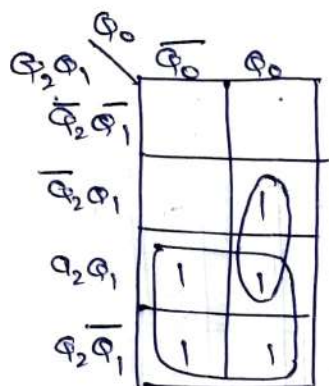
x	1	1	x
x	1	1	x
x	x	x	x
x	1	x	x

$$K_0 = 1$$

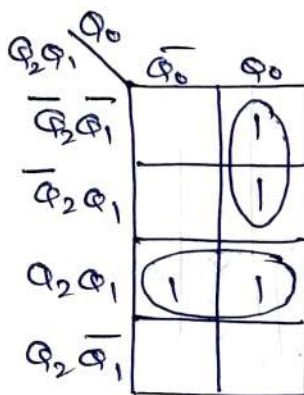


Design a modulo-5 synchronous counter using T-FF (Avoid Lock-out problem).

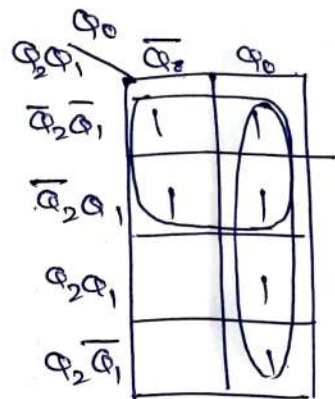
	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
2	0	1	0	0	1	1	0	0	1
3	0	1	1	1	0	0	1	1	1
4	1	0	0	0	0	0	1	0	0
5	1	0	1	0	0	0	1	0	1
6	1	1	0	0	0	0	1	1	0
7	1	1	1	0	0	0	1	1	1



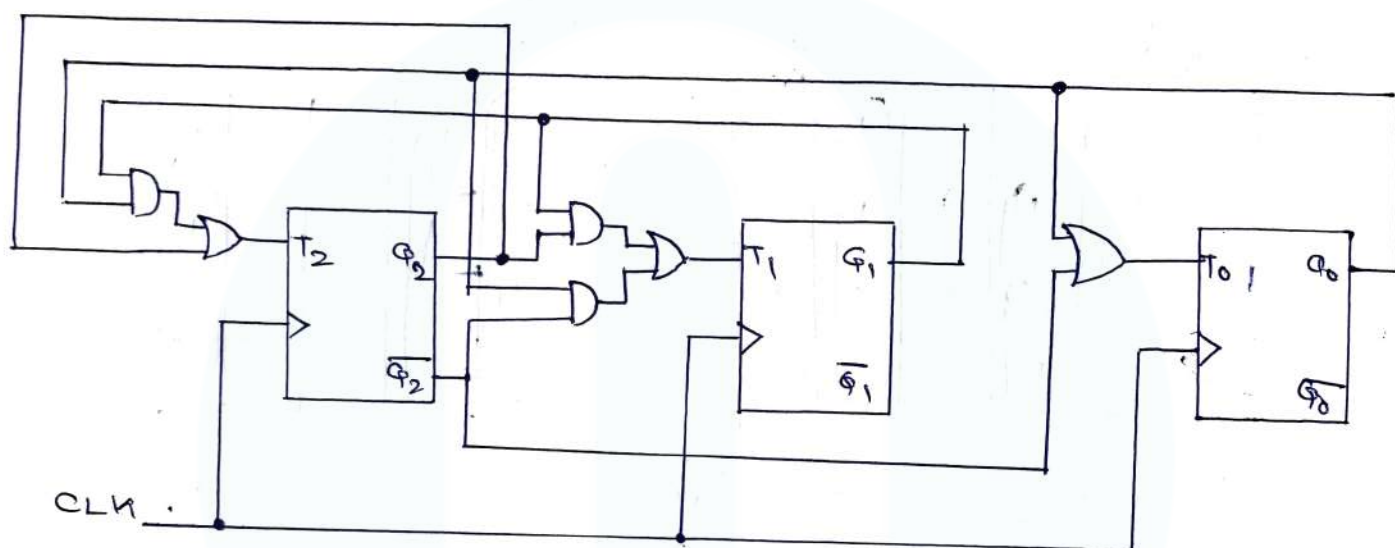
$$T_2 = Q_2 + Q_1 Q_0$$



$$T_1 = Q_2 Q_1 + Q_2 Q_0$$

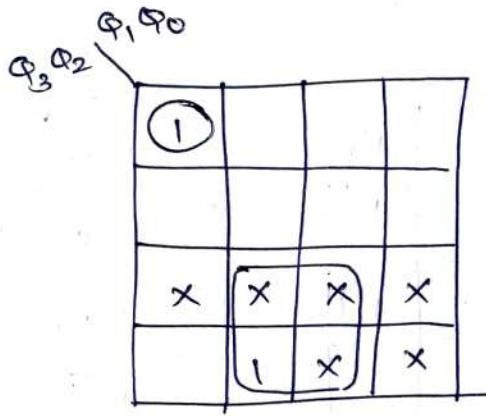


$$T_0 = Q_2 + Q_0$$

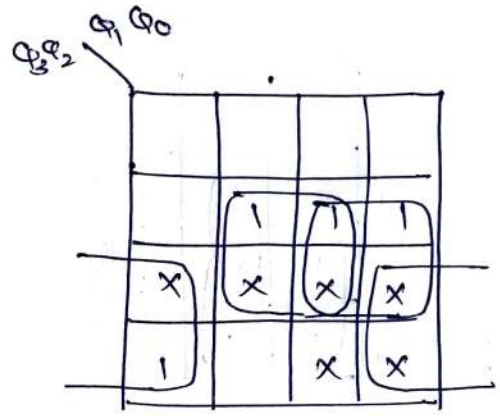


Design a synchronous down counter which count from 9 to 0 using D-FF.

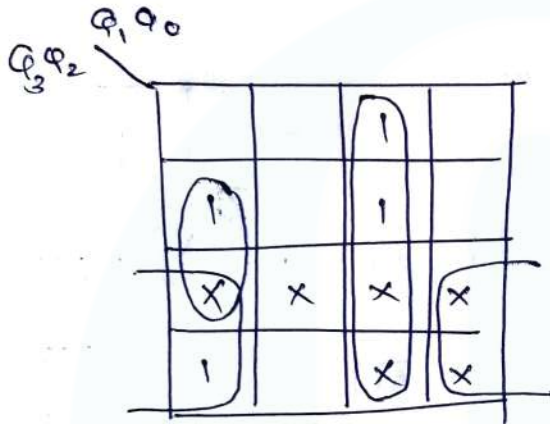
	Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	D_3	D_2	D_1	D_0
0	0	0	0	0	↓	0	0	1	1	0	0	1
1	0	0	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1	0	0	0	1	0
4	0	1	0	0	0	0	1	1	0	0	1	1
5	0	1	0	1	0	1	0	0	0	1	0	0
6	0	1	1	0	0	1	0	1	0	1	0	1
7	0	1	1	1	0	1	1	0	0	1	1	0
8	1	0	0	0	0	1	1	1	0	1	1	1
9	1	0	0	1	1	0	0	0	1	0	0	0



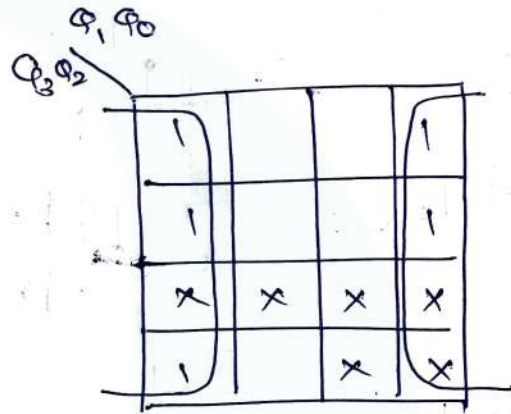
$$D_3 = \overline{Q_3} \overline{Q_2} \overline{Q_1} \overline{Q_0} + Q_3 Q_0$$



$$D_2 = Q_2 Q_1 + Q_2 Q_0 + Q_3 \overline{Q_0}$$



$$D_1 = Q_1 Q_0 + Q_3 \overline{Q_0} + Q_2 \overline{Q_1} \overline{Q_0}$$

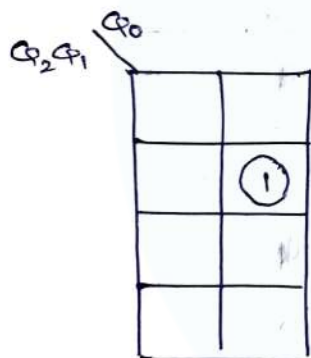
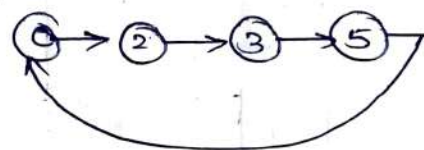


$$D_0 = \overline{Q_0}$$

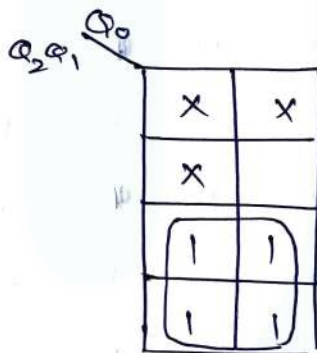
Design a circuit which generate a sequence

0, 2, 3, 5, 0, 2, 3, 5, ... using SR-FF.

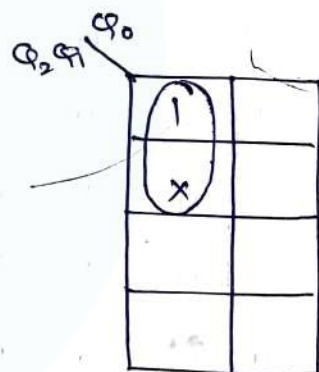
	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	S_2	R_2	S_1	R_1	S_0	R_0
0	0	0	0	0	1	0	0	X	0	0	0	X
1	0	0	1	0	0	0	0	X	1	X	0	1
2	0	1	0	0	1	1	0	X	X	0	1	0
3	0	1	1	1	0	1	1	0	0	1	X	0
4	1	0	0	0	0	0	0	1	0	X	0	X
5	1	0	1	0	0	0	0	1	0	X	0	1
6	1	1	0	0	0	0	0	1	0	1	0	X
7	1	1	1	0	0	0	0	1	0	1	0	1



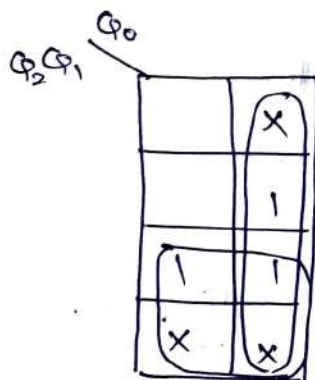
$$S_2 = \overline{Q_2} Q_1 Q_0$$



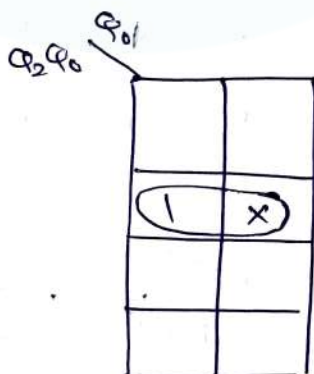
$$R_2 = Q_2$$



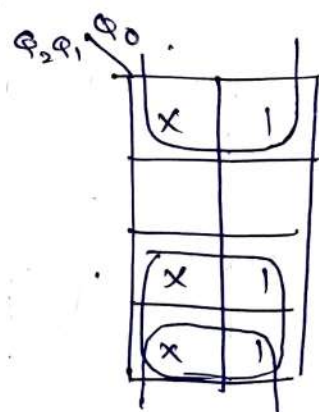
$$S_1 = \overline{Q_2} \overline{Q_0}$$



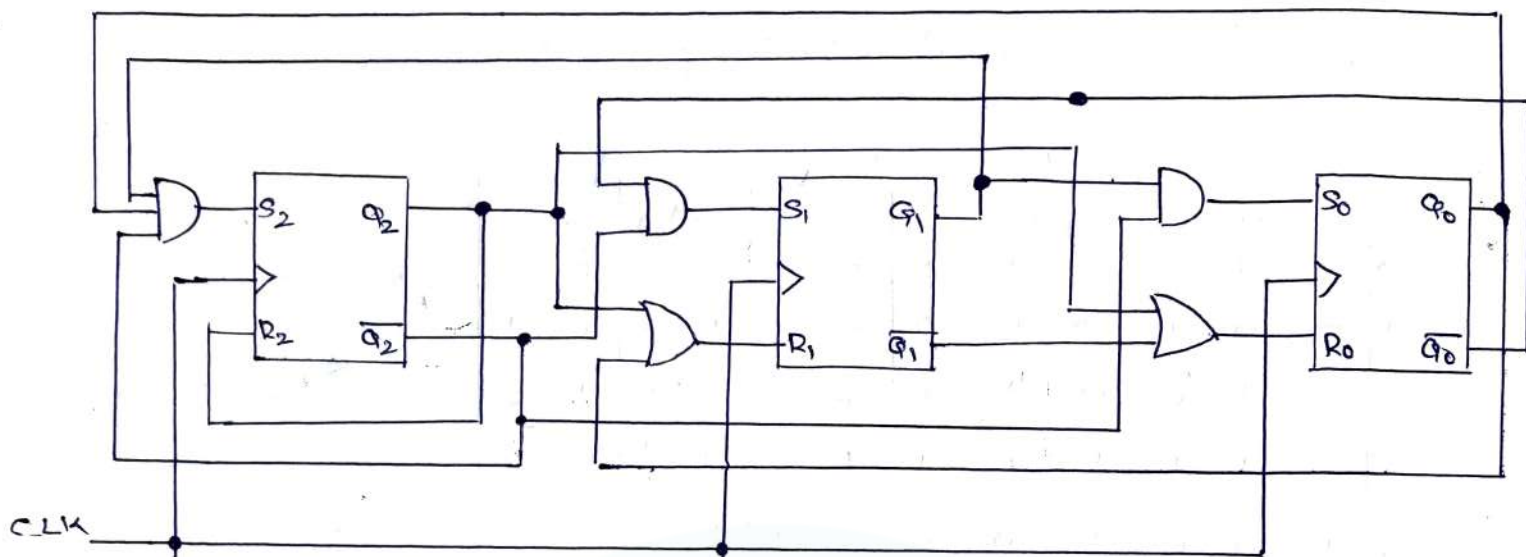
$$R_1 = Q_2 + Q_0$$



$$S_0 = \overline{Q_2} Q_1$$

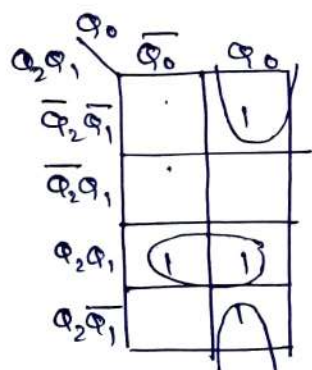


$$R_0 = Q_2 + \overline{Q_1}$$

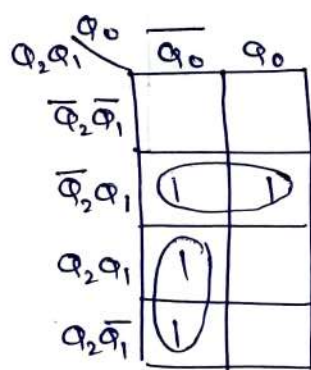


Design a circuit which generate a sequence 3, 1, 4, 7, 3, 1, 4, ... using T-FF.

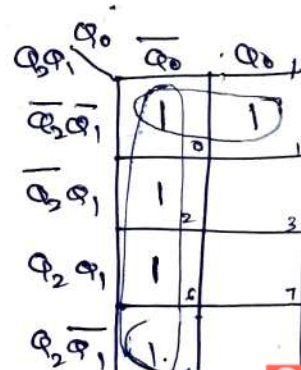
	Q_2	Q_1	Q_0	Q_2	Q_1	Q_0	T_2	T_1	T_0
0	0	0	0	0	0	0	0	0	1
1	0	0	1	1	0	0	1	0	1
2	0	1	0	0	0	1	0	1	1
3	0	1	1	0	0	1	0	1	0
4	1	0	0	1	1	1	0	1	1
5	1	0	1	0	0	1	1	0	0
6	1	1	0	0	0	1	1	1	1
7	1	1	1	0	1	1	1	0	0



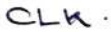
$$T_2 = \overline{Q_1}Q_0 + Q_2Q_1$$



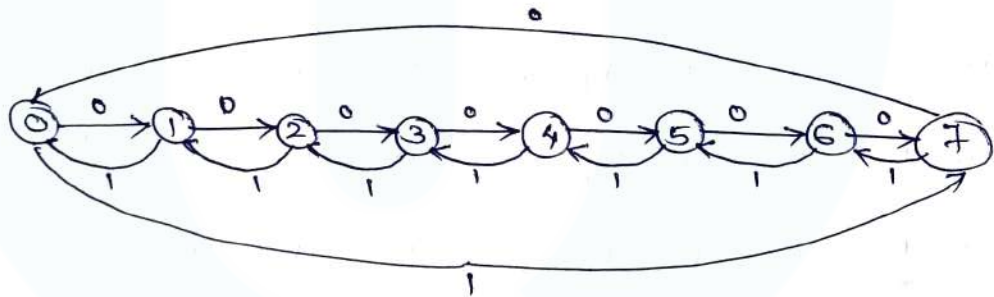
$$T_1 = \overline{Q_2}Q_1 + Q_2Q_0$$



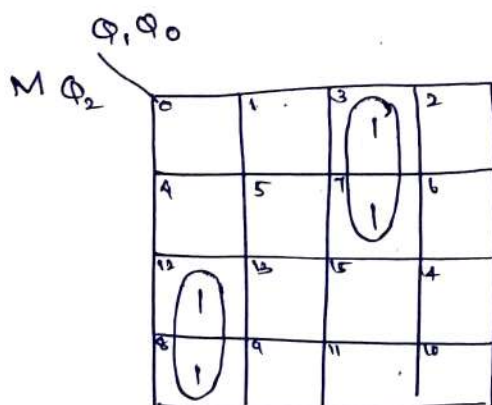
$$T_0 = \overline{Q_0} + Q_2Q_1$$



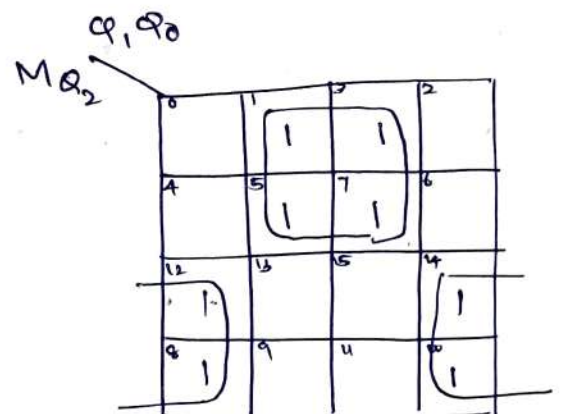
If $M=0$, counter counts upwards and if $M=1$ counter counts downwards.



	M	Q ₂	Q ₁	Q ₀	Q ₂	Q ₁	Q ₀	T ₂	T ₁	T ₀
0	0	0	0	0	0	0	1	0	0	1
1	0	0	0	1	0	1	0	0	1	1
2	0	0	1	0	0	1	1	0	0	1
3	0	0	1	1	1	0	0	1	1	1
4	0	1	0	0	1	0	1	0	0	1
5	0	1	0	1	1	1	0	0	1	1
6	0	1	1	0	1	1	1	0	0	1
7	0	1	1	1	0	0	0	1	1	1
15	1	1	1	1	0	1	0	0	0	1
14	1	1	1	0	1	0	1	0	1	1
13	1	1	0	1	1	0	0	0	0	1
12	1	1	0	0	0	1	1	1	1	1
11	1	0	1	1	0	1	0	0	0	1
10	1	0	1	0	0	0	1	0	1	1
9	1	0	0	1	0	0	0	0	0	1
8	1	0	0	0	1	1	1	1	1	1



$$T_2 = \overline{M} Q_1 Q_0 + M \overline{Q}_1 \overline{Q}_0$$



$$T_1 = \overline{M} Q_0 + M \overline{Q}_0$$

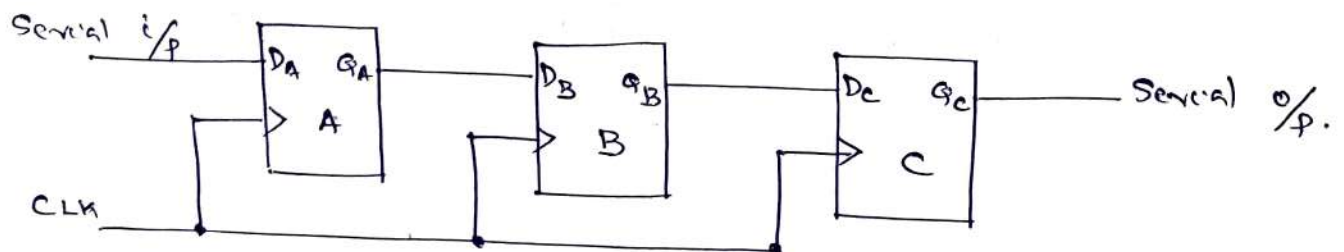
SHIFT REGISTERS.

A FF is a memory element or storage cell that can store one bit of information. A series of such FF connected in cascade can store several bits of data. Such a storage cell is known as Registers. A register that can be used to store and retrieve data by shifting is called shift Registers. The basic FF used in shift register is D-FF.

Types of Shift Registers,

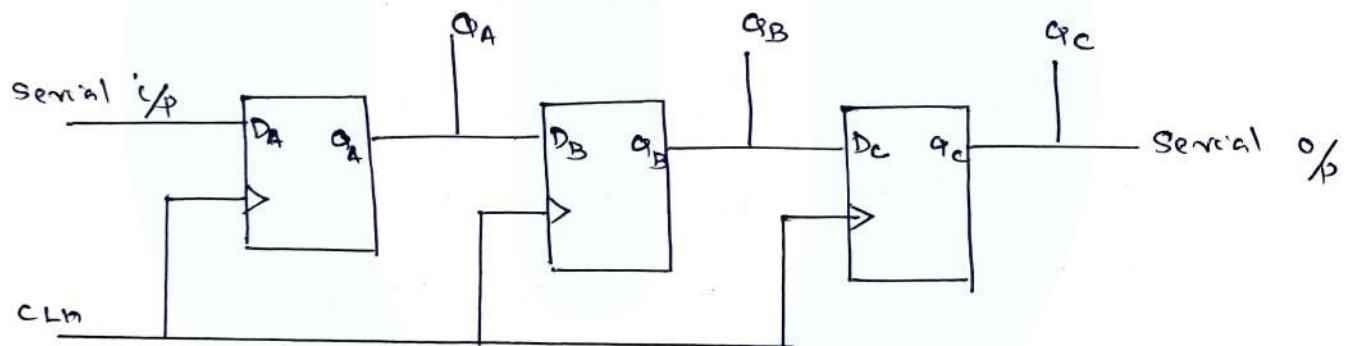
- (1) Serial i/p Serial o/p (SISO).
- (2) Serial i/p Parallel o/p (SIPO)
- (3) Parallel i/p Serial o/p (PISO)
- (4) Parallel i/p Parallel o/p (PIPO).

SISO

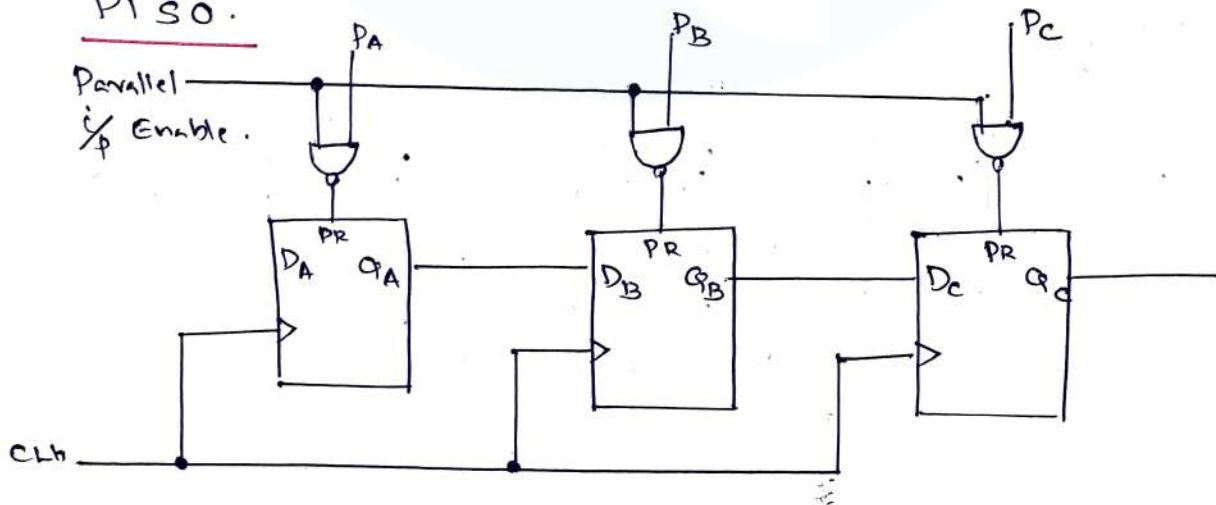


CLK.	D_A	Q_A	Q_B	Q_C
0th	1	0	0	0
1st	0	1	0	0
2nd	1	0	1	0
3rd	1	1	0	1
4th	0	1	1	0
5th	0	0	1	1
6th	0	0	0	1
7th	0	0	0	0

SISO.

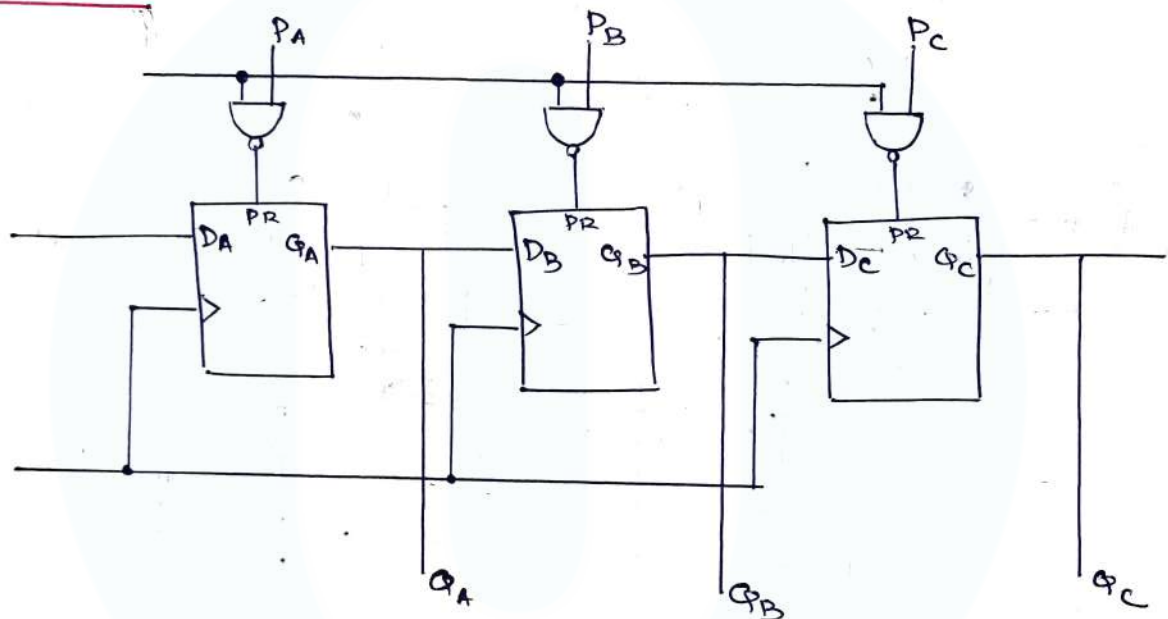


PISO.

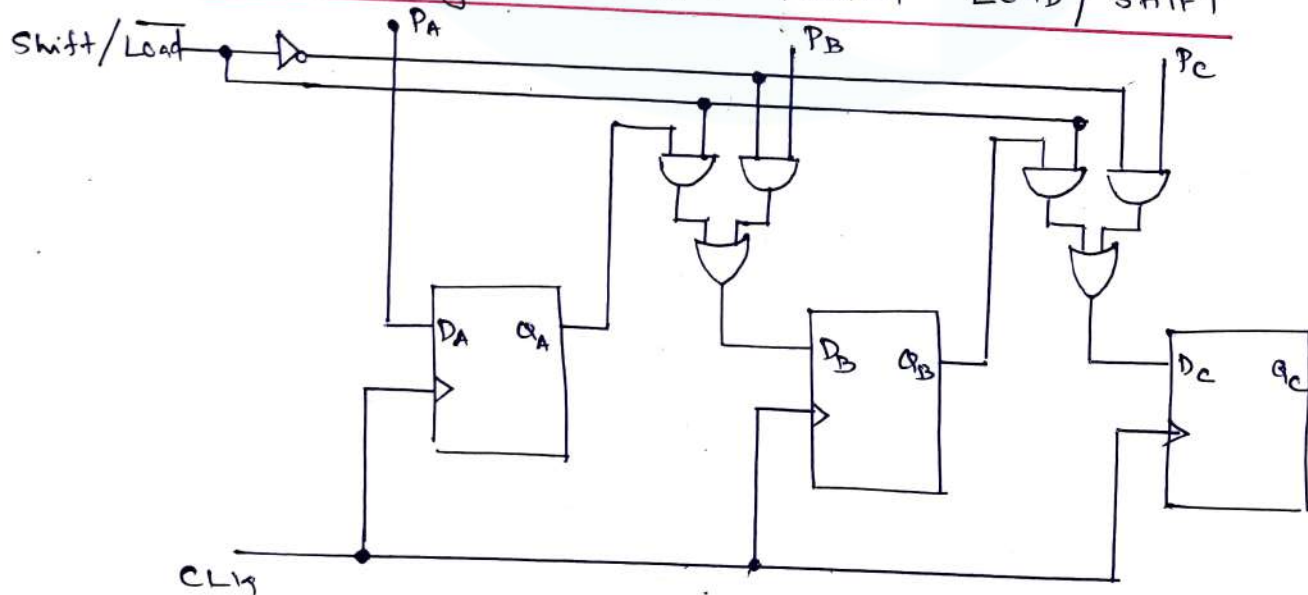


To store the bit sequenu 101 in a 11el
 i/p shift register, apply the bit sequenu to the i/p
 line P_A, P_B, P_C and make the 11el i/p enable line
 high. Now the preset i/p's (PR) of the FFs
 becomes 010 and therefore the o/p's Q_A, Q_B, Q_C
 becomes 101 respectively

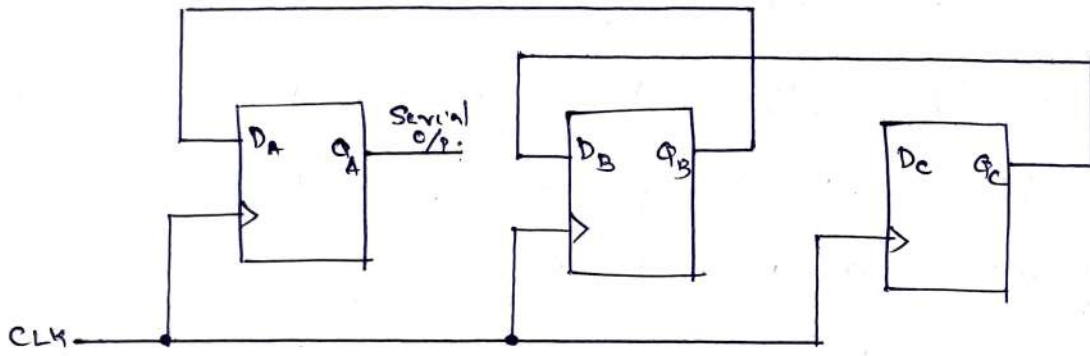
PIPO.



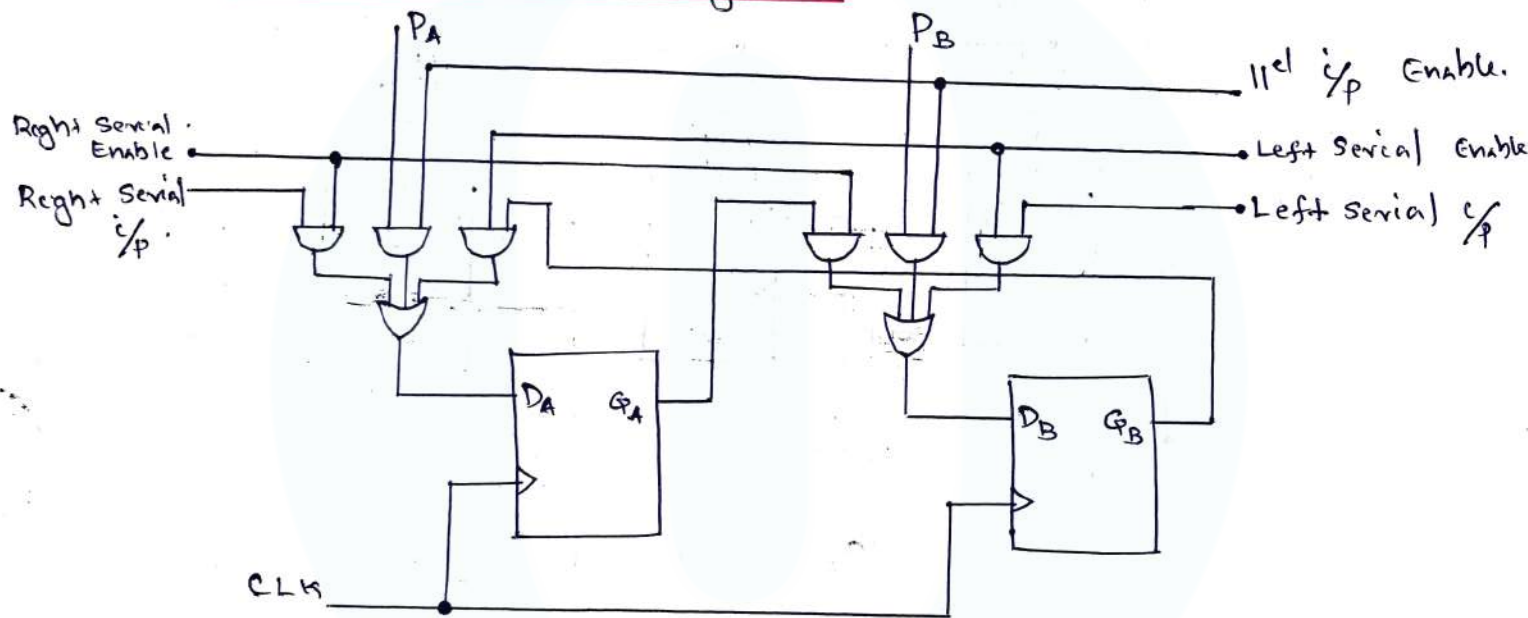
Shift Registers with Parallel LOAD / SHIFT



Left Shift Register.



Universal Shift Register.



APPLICATIONS OF SHIFT REGISTER.

1. As a delay line.

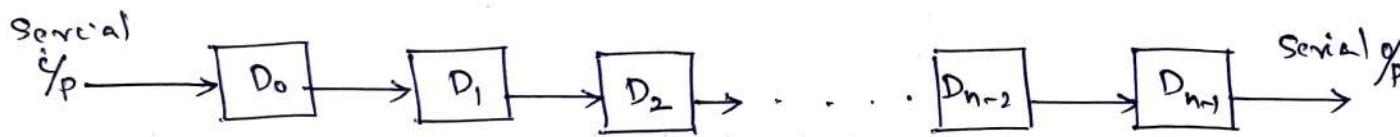
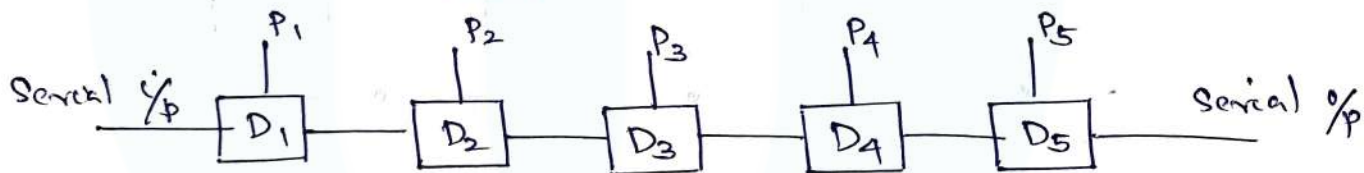


Figure shows n D FF connected in series. The propagation delay of each FF is t_p ns. Thus the total time delay is $n t_p$. Therefore, the ckt can be used as a delay line to produce a required amount of time delay.

2. Sequence Generator.

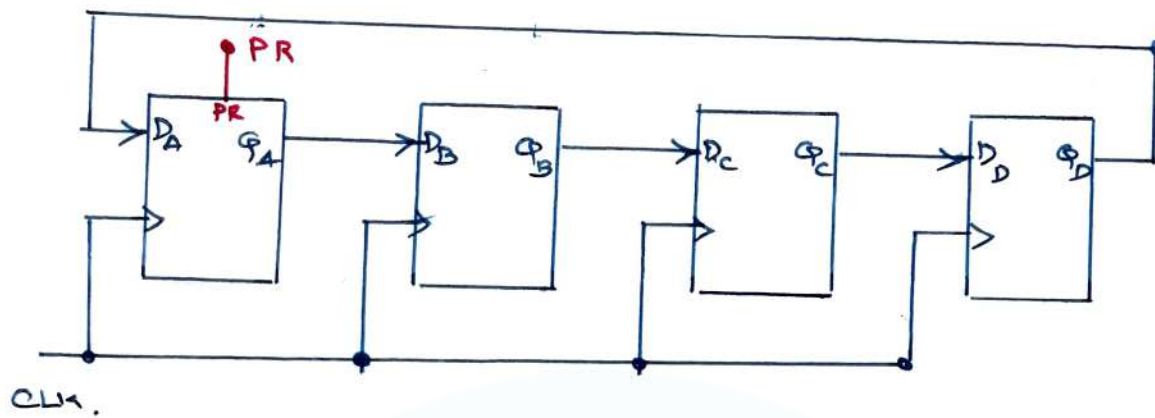


When $CLK=1$, the data applied to the n i/p terminals will appear at the respective o/p's of the FF. On applying the clock pulse we get the data serially at the o/p terminal of the last FF. Any desired pattern can be generated by using this type of shift register ckt.

3. Ring Counter.

A shift register can be connected in series

RING COUNTER.



- we introduce a 1 as i/x at the 1st stage, & we can circulate this bit 1 by successive shift of CLK pulse through individual FFs.

CLK	Q_A	Q_B	Q_C	Q_D
	0	0	0	0
1	1	0	0	0 ✓
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0

- For a 4-bit ring counter, the continuous sequence repeated after 4 clock pulses

- In general for an n-bit ring counter the counting sequence repeated after n CLK pulses.

From equ. (20) in the previous section

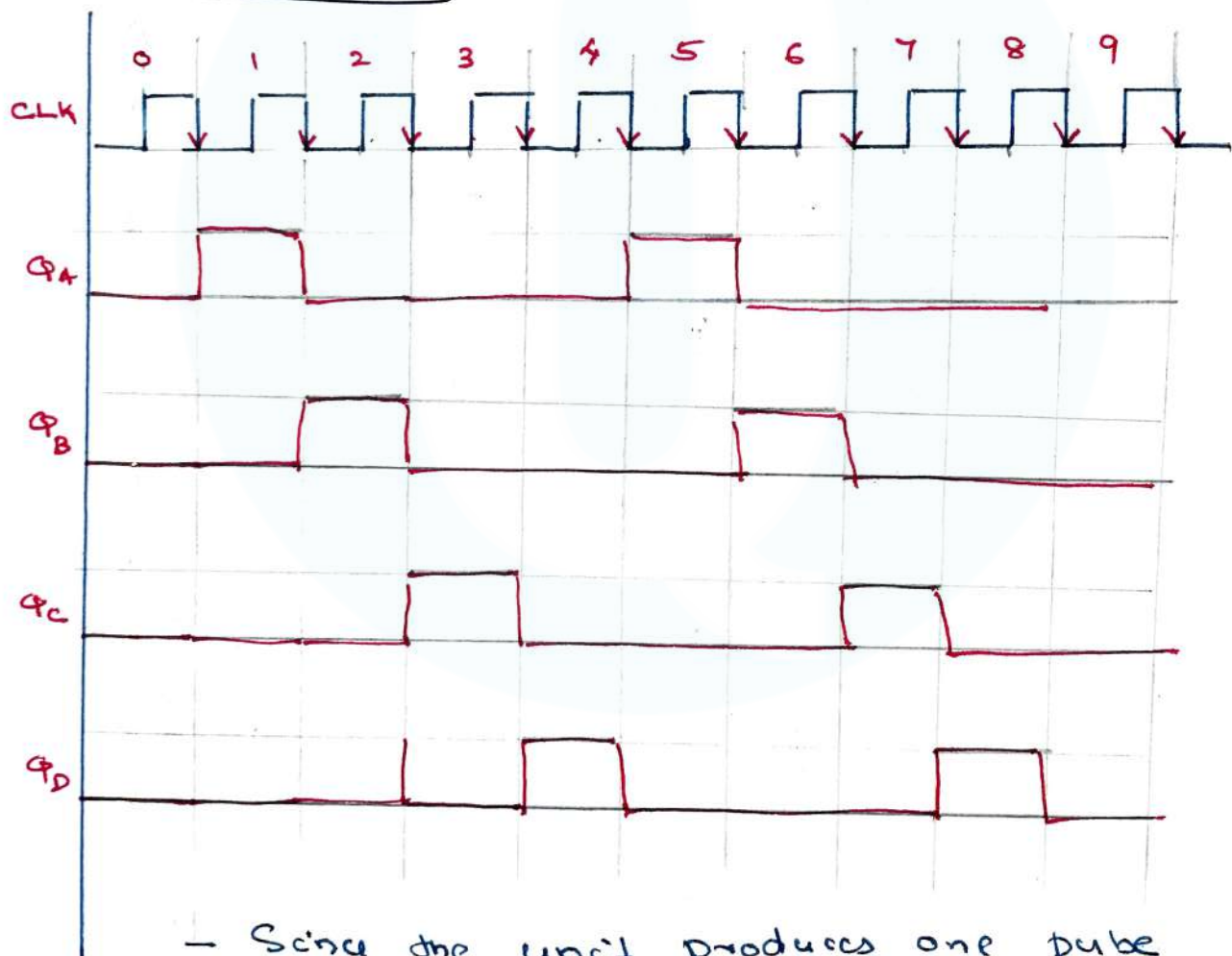
$$E_x = -\frac{1}{h^2} \frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial y} \quad \text{--- (20)(a)}$$

$$E_y = -\frac{1}{h^2} \frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2} \frac{\partial H_z}{\partial x} \quad \text{--- (20)(b)}$$

$$H_x = -\frac{1}{h^2} \frac{\partial H_z}{\partial x} + \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial y} \quad \text{--- (20)(c)}$$

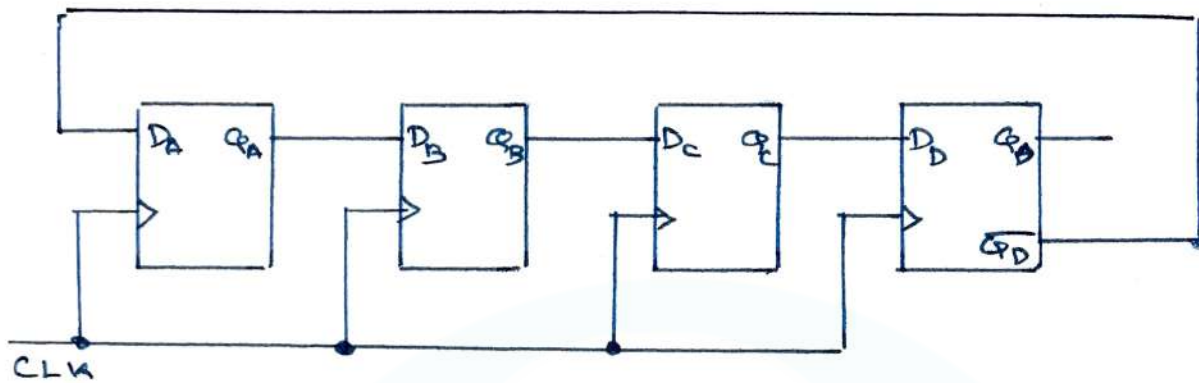
$$H_y = -\frac{1}{h^2} \frac{\partial H_z}{\partial y} - \frac{j\omega\epsilon}{h^2} \frac{\partial E_z}{\partial x} \quad \text{--- (20)(d)}$$

TIMING DIAGRAM



— Since the unit produces one pulse for every N -i/p clock pulses, this can be used as an $N:1$ divider.

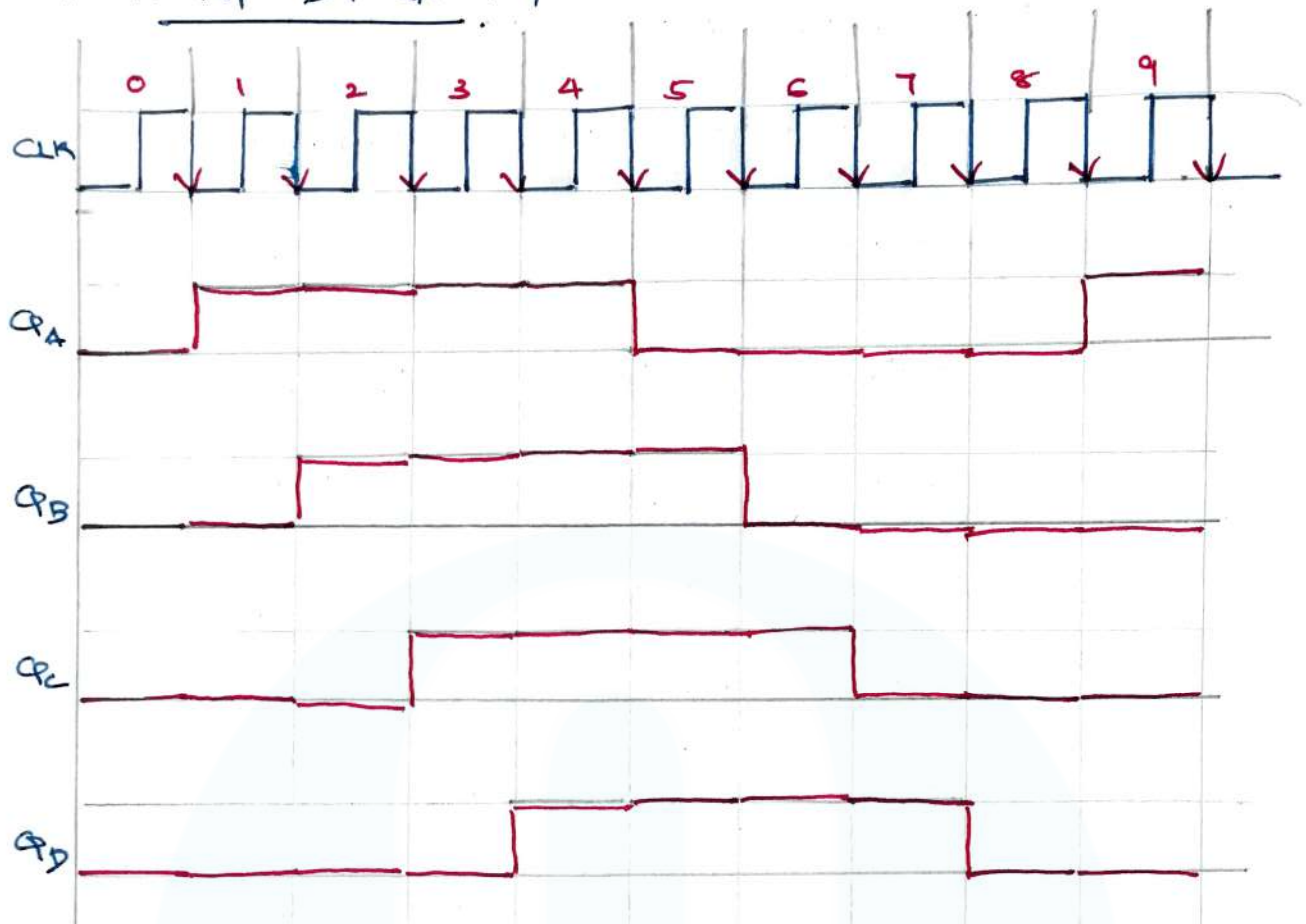
JOHNSON COUNTER / TWISTED RING COUNTER / SWITCHED TAIL COUNTER



- Initially all the FFs cleared using clear ϕ 's.
- For 4-bit Johnson counter the counting sequence repeats after 8-clock pulses.
- In general for an n -bit Johnson counter the counting sequence repeats after $2n$ clk pulses.

CLK	Q_A	Q_B	Q_C	Q_D	
0	0	0	0	0	$\overline{Q_D} = 1$
1	1	0	0	0	
2	1	1	0	0	
3	1	1	1	0	
4	1	1	1	1	$\overline{Q_D} = 0$
5	0	1	1	1	
6	0	0	1	1	
7	0	0	0	1	
8	0	0	0	0	

TIMING DIAGRAM



- Since the unit produces one pulse for every $2N$ $\frac{1}{p}$ clock pulses (Here 8), this can be used as an $2N:1$ divider.